

How is life sustained in an ecosystem

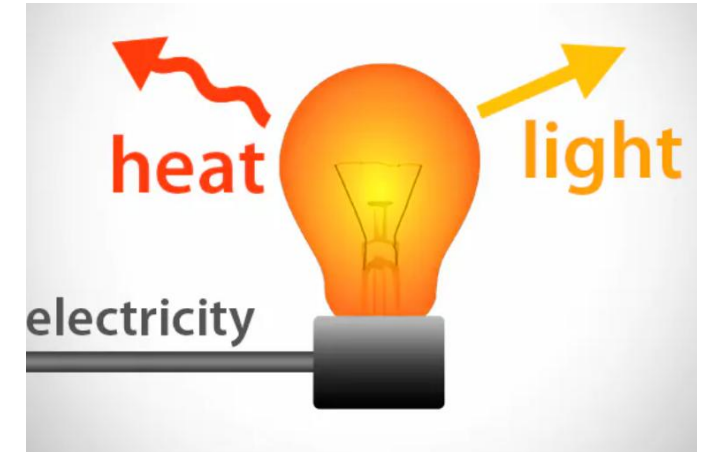
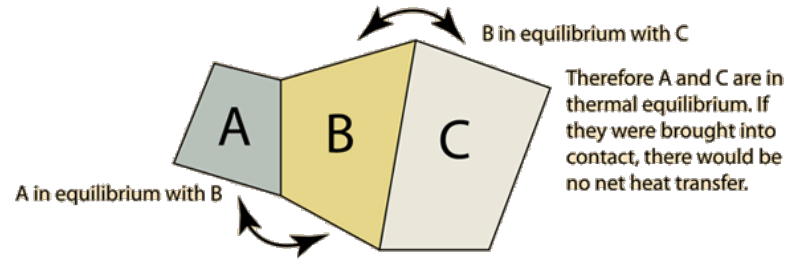
- It sustains life systems and ecological processes by energy flow
- Synthesis of organic matter by producers, i.e, productivity
- Decomposition of dead organic matter
- Cycling of minerals and nutrients



Energy is needed for us to do all kinds of work.



Thermodynamic laws provide the framework for bioenergetics



**0. Thermal
Equilibrium**



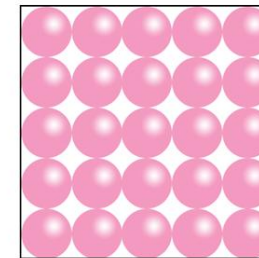
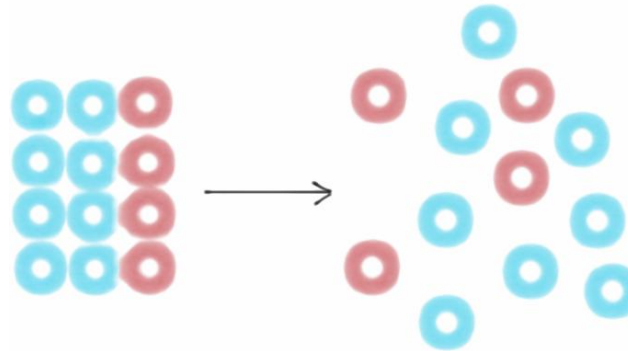
**1. Energy
conservation**



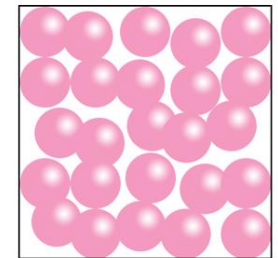
**2. ENTROPY
expansion**



**3. Absolute
temperature**



$T = 0$
 $S = 0$



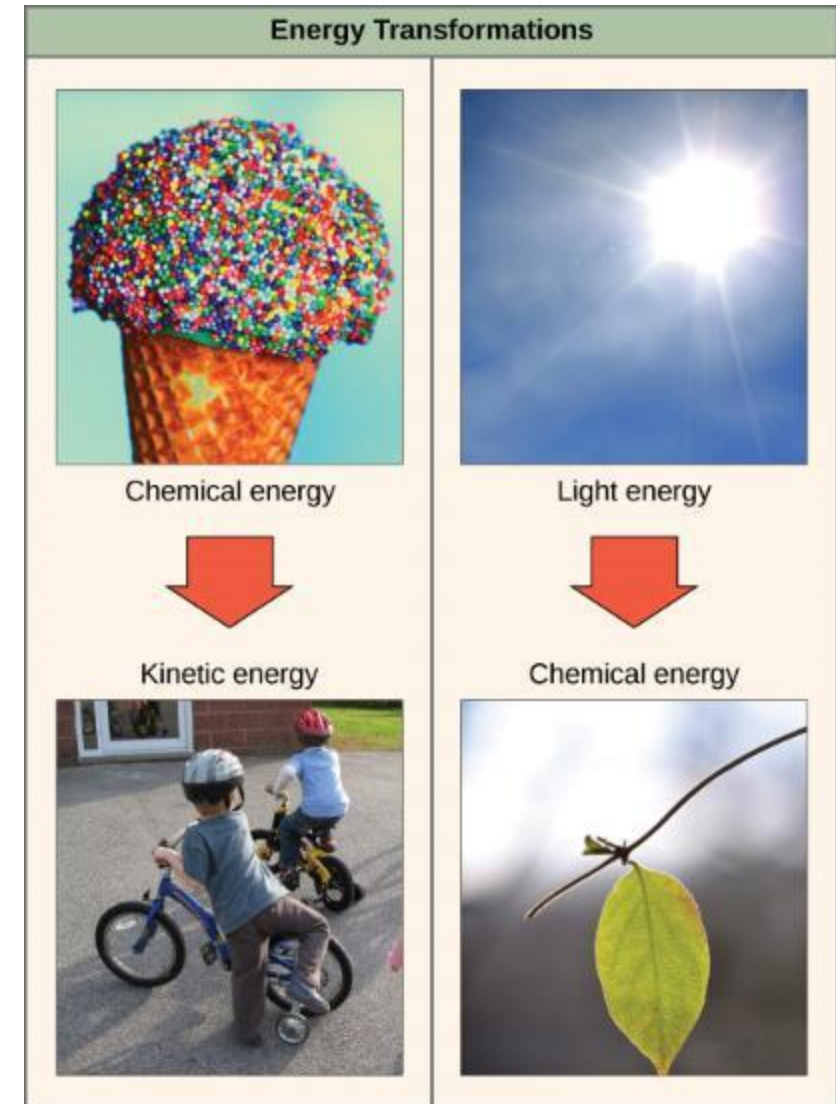
$T > 0$
 $S > 0$

Energy transformations

Cellular energy can be *potential* or *kinetic*.

Potential energy is stationary, and it exists as chemical bonds, concentration gradients, and electrical charges in the cells.

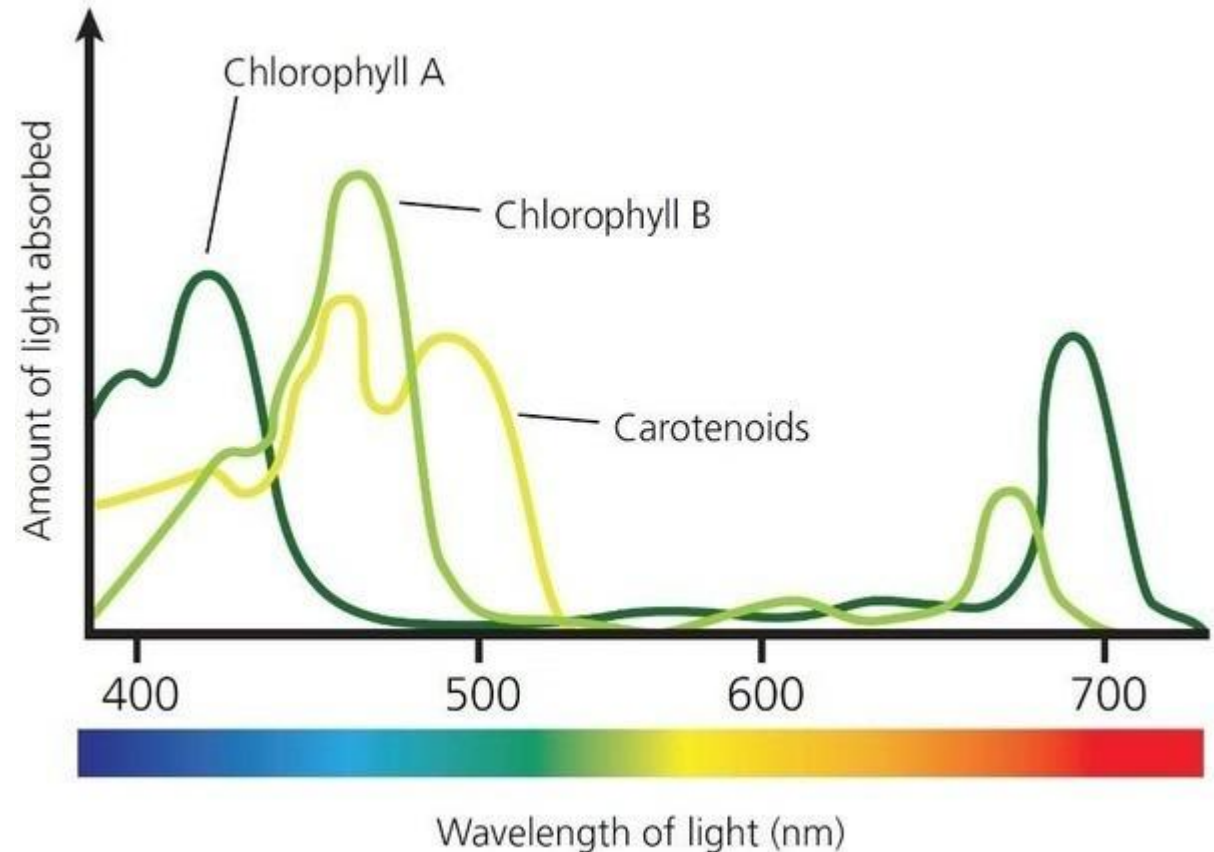
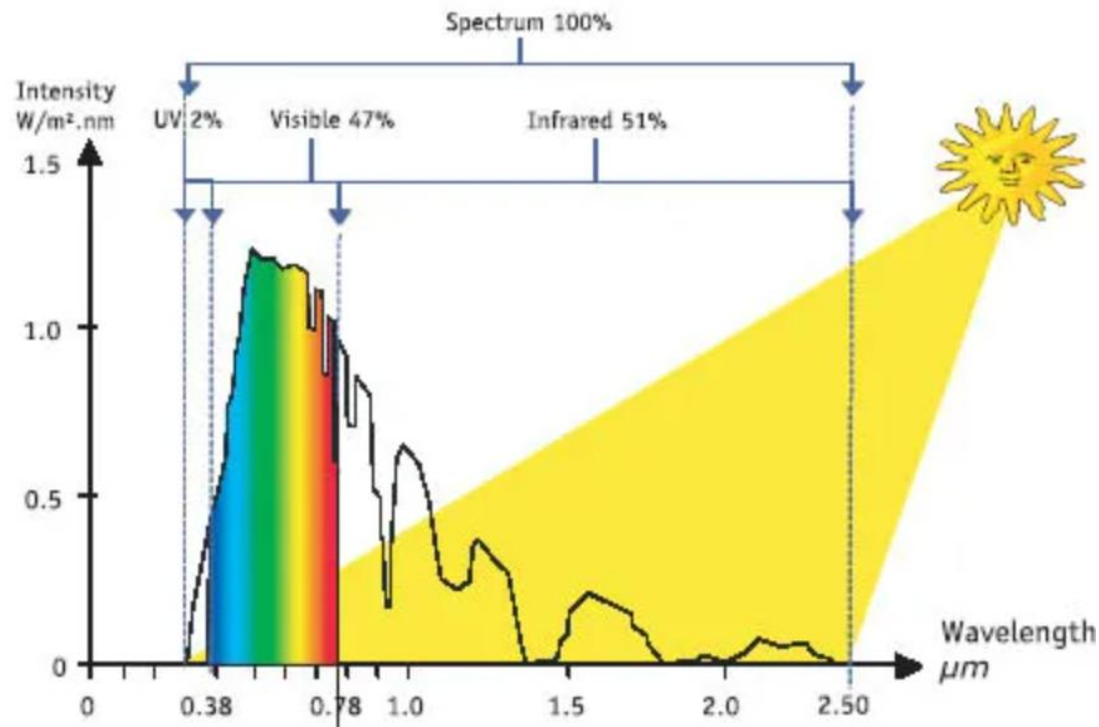
Kinetic energy is energy in motion, which results in the movement of molecules. Heat or thermal energy, for example, causes molecules to oscillate. In the cells, kinetic energy is converted from potential energy, moving electrons, hydrogen ions, and other charged particles



Source of energy

- Less than 50 percent of the incident solar radiation is photosynthetically active radiation
- Plants and photosynthetic bacteria capture 2-10 % of the photosynthetic active radiation and this small amount of energy sustains the entire living world.

SOLAR SPECTRUM



Fundamental laws governing energy flow

First law of thermodynamics

It states that energy can neither be created nor be destroyed. It can only be transferred from one form to another.

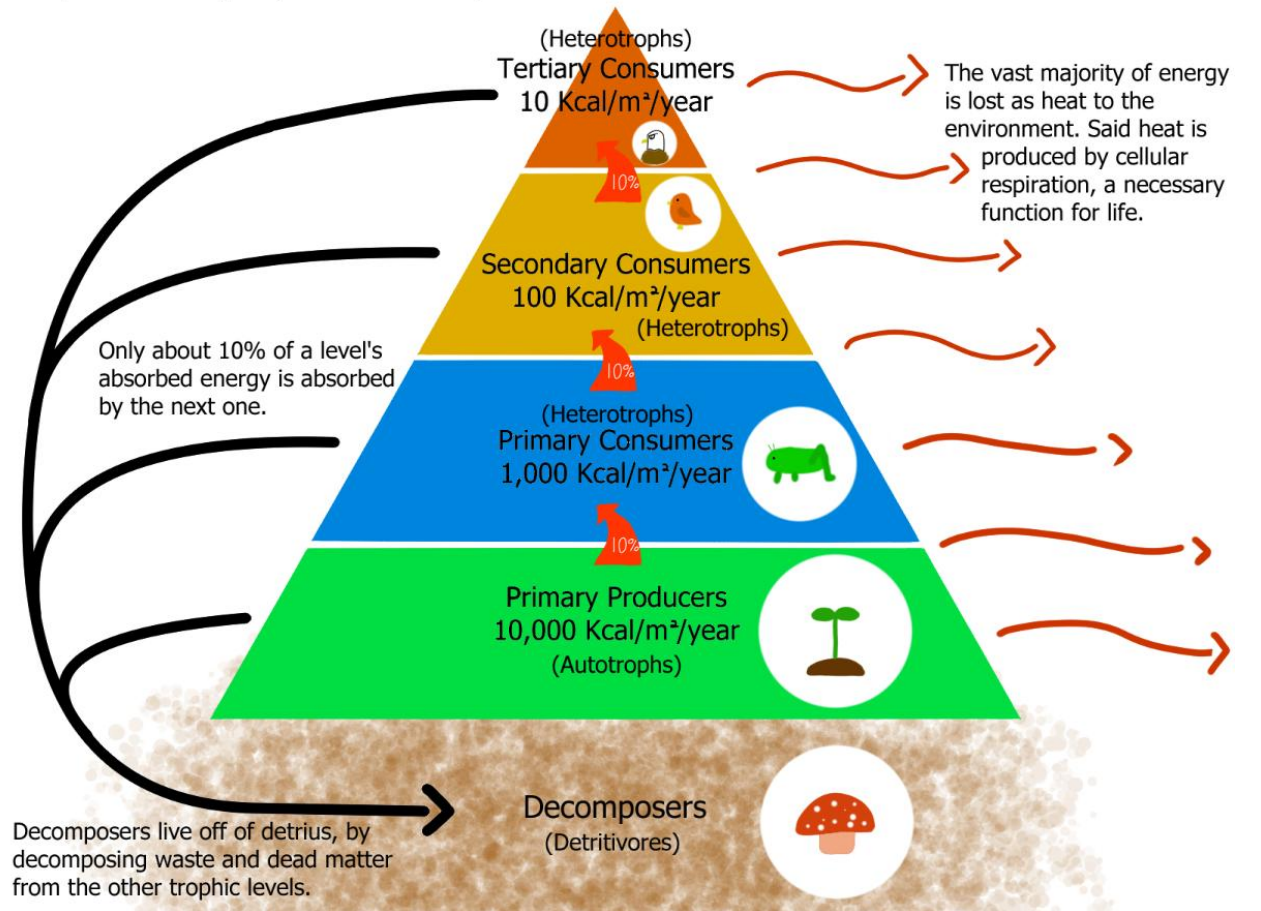
Hence the total energy of the universe is constant. During photosynthesis, there is a transformation of light energy (sunlight) into chemical energy (food).

Second law of thermodynamics

The energy transformation is not hundred percent efficient. So, some energy is dissipated as heat energy into the surroundings, while transforming from one form to another. Hence while energy gets transferred from one trophic level to another, some part of the energy is lost as heat, which cannot be reused. Therefore, energy flow in an ecosystem is unidirectional or one way flow from producers through herbivores to carnivores. So the energy can be used only once and cannot be recycled.

Trophic Levels & Energy Transfer

Trophic levels are split by a who-eats-who system.



The energy available for the individuals at higher trophic levels will be gradually declining. Hence after 5th trophic level, the available energy will be insufficient for survival. This is the reason why the trophic levels are limited.

Bioenergetics

- Describes the transfer and utilization of energy in biologic systems. It makes use of a few basic ideas from the field of thermo - dynamics, particularly the concept of free energy.
- Changes in free energy (ΔG) provide a measure of the energetic feasibility of a chemical reaction and can, therefore, allow prediction of whether a reaction or process can take place.
- In short, bio - energetics predicts if a process is possible, whereas kinetics measures how fast the reaction occurs

FREE ENERGY

Direction and extent to which a chemical reaction proceeds is determined by the degree to which two factors change during the reaction.

ΔG , represents the change in free energy at any specified concentration of products and reactants.

The standard free energy change, ΔG° is the energy change when reactants and products are at a concentration of 1 mol/L. [Note: The concentration of protons is assumed to be 10^{-7} mol/L, that is, pH = 7.] Furthermore, ΔG° can readily be determined from measurement of the equilibrium constant

ΔG : CHANGE IN FREE ENERGY

- Energy available to do work.
- Approaches zero as reaction proceeds to equilibrium.
- Predicts whether a reaction is favorable.

ΔH : CHANGE IN ENTHALPY

- Heat released or absorbed during a reaction.
- Does not predict whether a reaction is favorable.

$$\Delta G = \Delta H - T\Delta S$$

ΔS : CHANGE IN ENTROPY

- Measure of randomness.
- Does not predict whether a reaction is favorable.

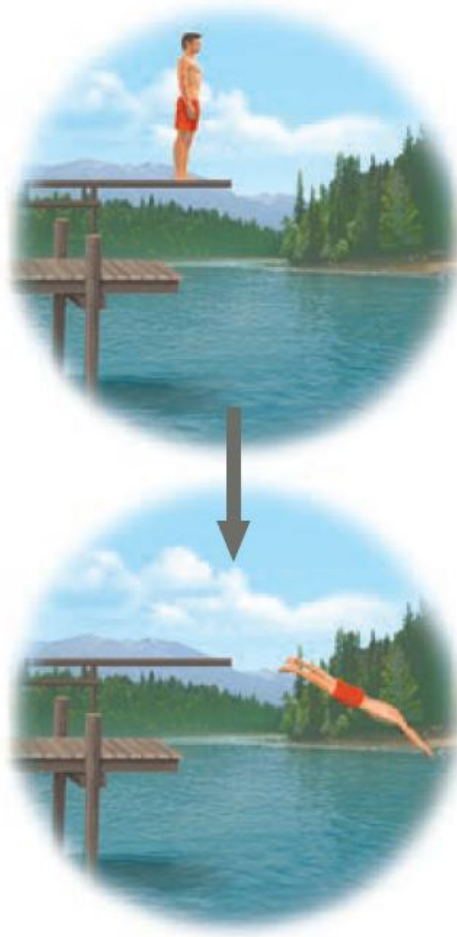
▼ **Figure 8.5 The relationship of free energy to stability, work capacity, and spontaneous change.** Unstable systems (top) are rich in free energy, G . They have a tendency to change spontaneously to a more stable state (bottom), and it is possible to harness this “downhill” change to perform work.

- More free energy (higher G)
- Less stable
- Greater work capacity

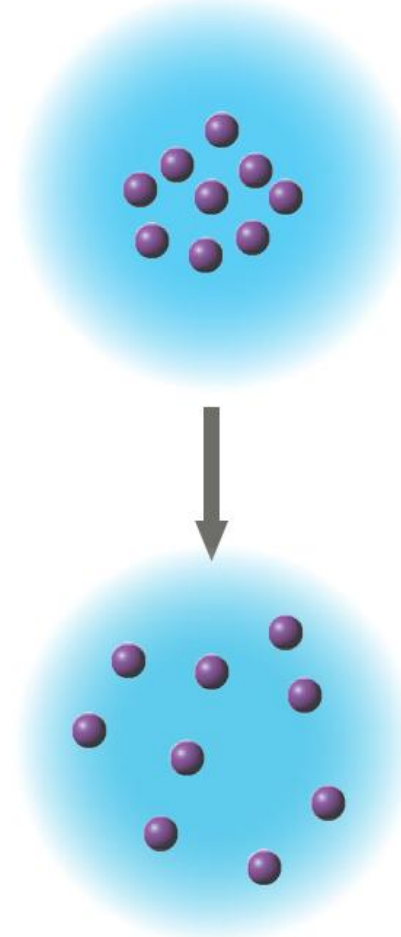
In a **spontaneous change**

- The free energy of the system decreases ($\Delta G < 0$)
- The system becomes more stable
- The released free energy can be harnessed to do work

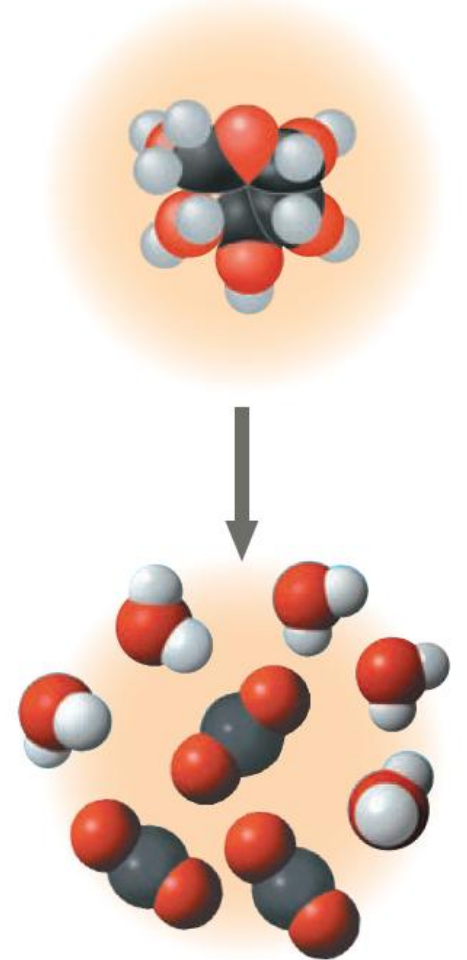
- Less free energy (lower G)
- More stable
- Less work capacity



(a) Gravitational motion. Objects move spontaneously from a higher altitude to a lower one.



(b) Diffusion. Molecules in a drop of dye diffuse until they are randomly dispersed.

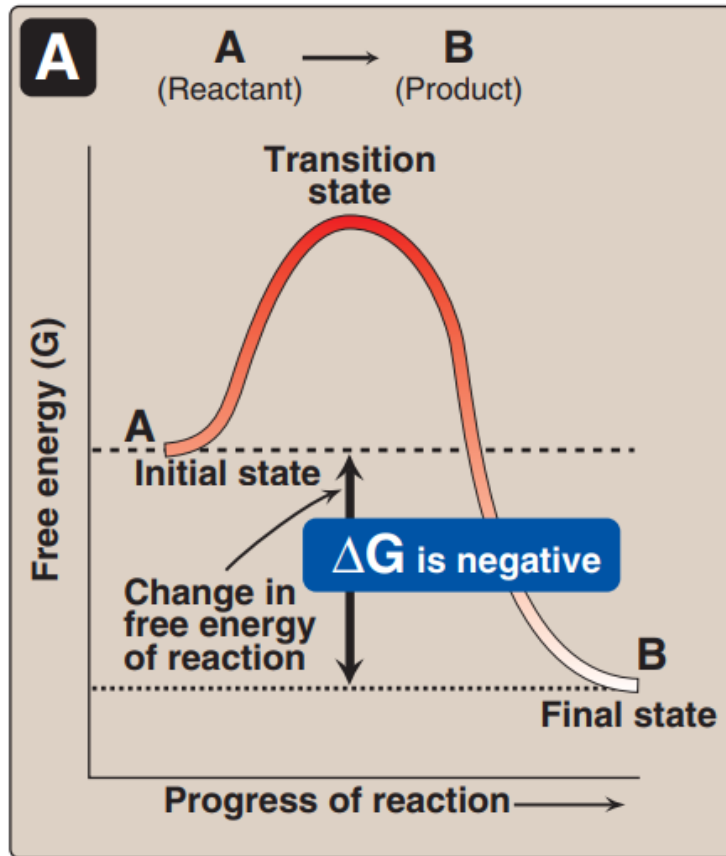


(c) Chemical reaction. In a cell, a glucose molecule is broken down into simpler molecules.

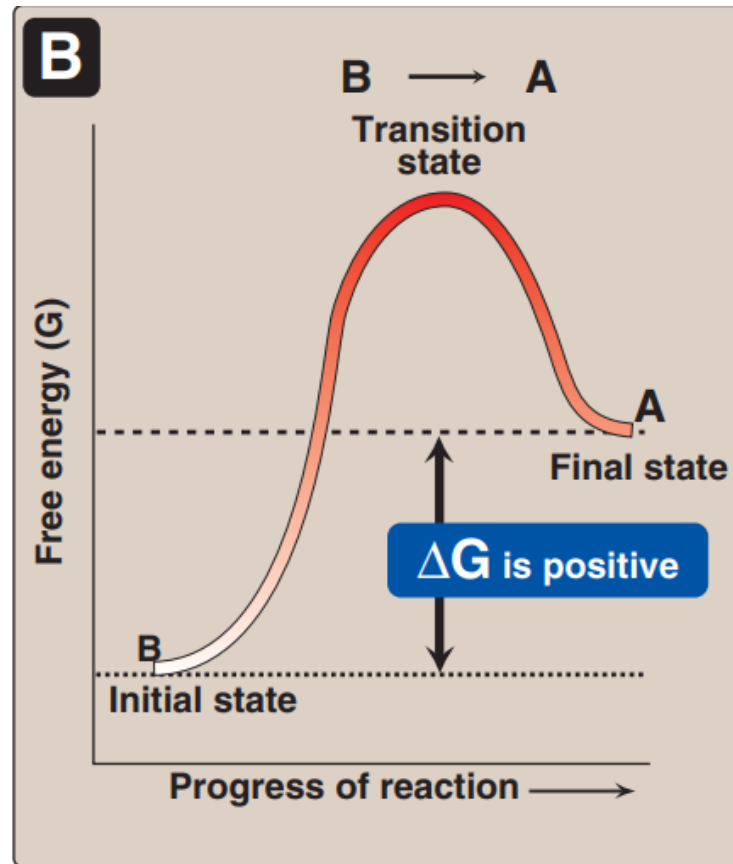
FREE ENERGY CHANGE

If $\Delta G = 0$, the reactants are in equilibrium

exergonic



endergonic

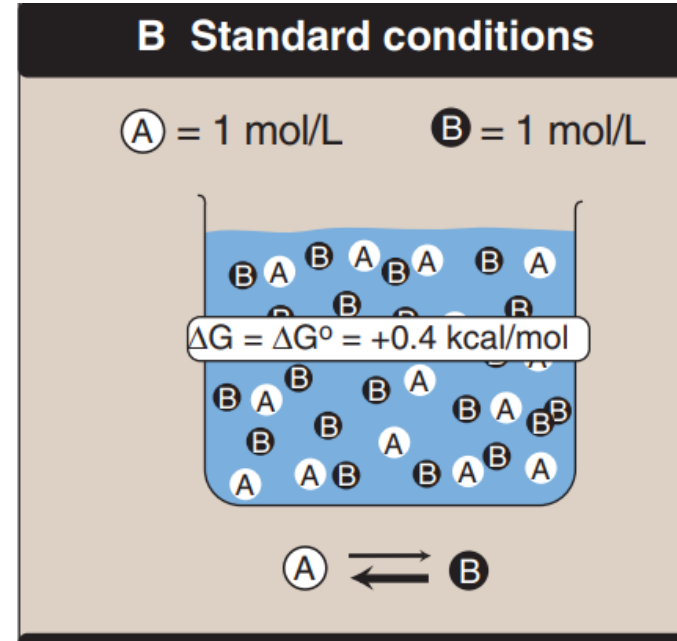
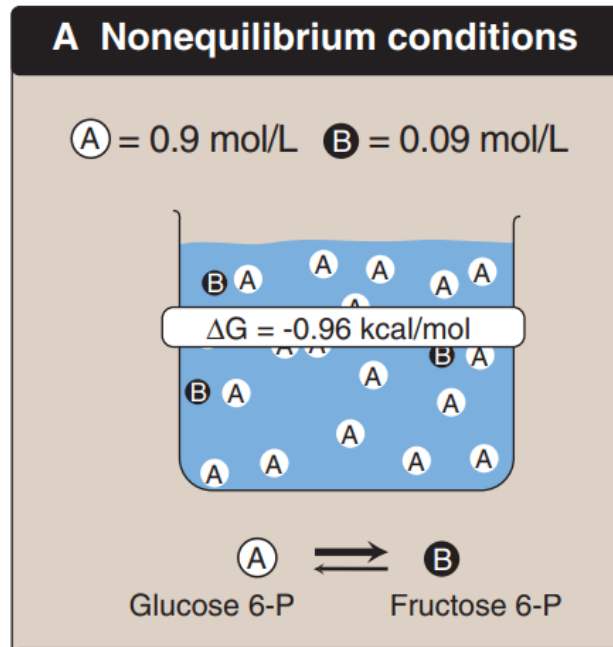


The free energy of the forward reaction ($A \rightarrow B$) is equal in magnitude but opposite in sign to that of the back reaction ($B \rightarrow A$)

$$\Delta G = \Delta G^\circ + RT \ln \frac{[B]}{[A]}$$

where ΔG° is the standard free energy change (see below)
R is the gas constant (1.987 cal/mol · degree)
T is the absolute temperature (°K)
[A] and [B] are the actual concentrations of the reactant and product
ln represents the natural logarithm

A reaction with a positive ΔG° can proceed in the forward direction (have a negative overall ΔG) if the ratio of products to reactants ($[B]/[A]$) is sufficiently small (that is, the ratio of reactants to products is large).

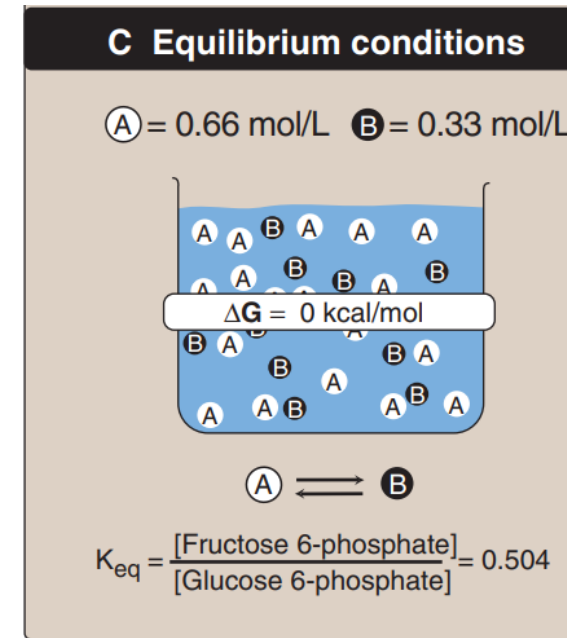


In a reaction $A \rightarrow B$, a point of equilibrium is reached at which no further net chemical change takes place—that is, when A is being converted to B as fast as B is being converted to A. In this state, the ratio of [B] to [A] is constant, regardless of the actual concentrations of the two compounds:

$$K_{eq} = \frac{[B]_{eq}}{[A]_{eq}}$$

$$\Delta G^{\circ} = -RT \ln K_{eq}$$

$$\Delta G = 0 = \Delta G^{\circ} + RT \ln \frac{[B]_{eq}}{[A]_{eq}}$$



This equation allows some simple predictions:

If $K_{eq} = 1$, then $\Delta G^{\circ} = 0$



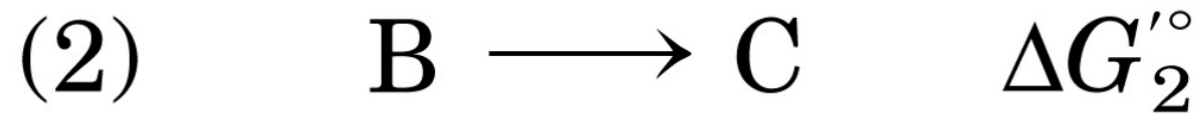
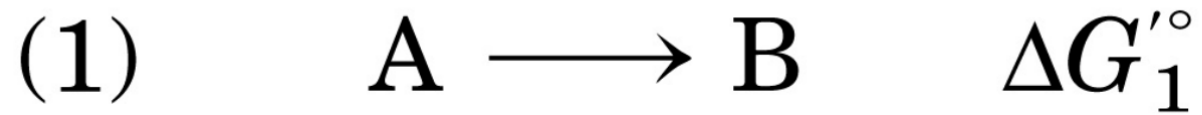
If $K_{eq} > 1$, then $\Delta G^{\circ} < 0$



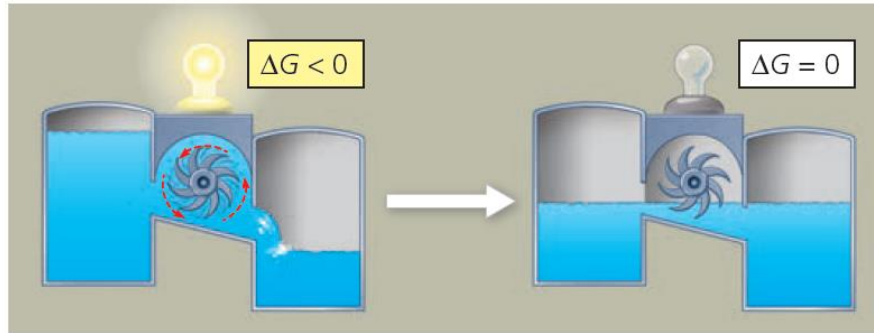
If $K_{eq} < 1$, then $\Delta G^{\circ} > 0$



Free energies are additive, thus a favorable reaction ($\Delta G_1 < 0$) can drive an unfavorable reaction ($\Delta G_2 > 0$), when $\Delta G_1 + \Delta G_2 < 0$



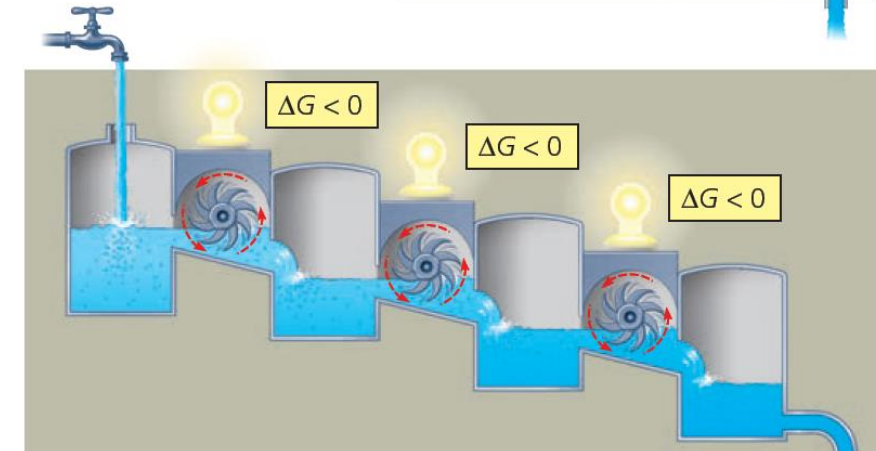
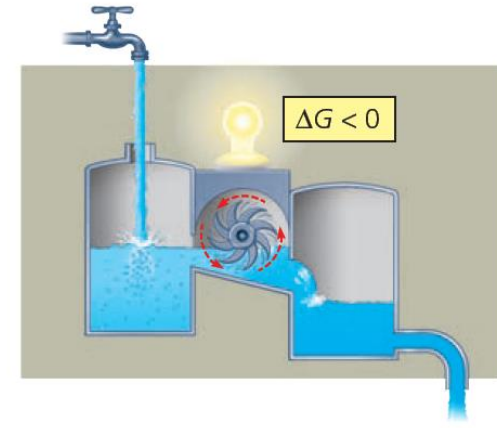
▼ **Figure 8.7 Equilibrium and work in an isolated hydroelectric system.** Water flowing downhill turns a turbine that drives a generator providing electricity to a lightbulb, but only until the system reaches equilibrium.



Reactions in an isolated system eventually reach equilibrium and can then do no work
 Because systems at equilibrium are at a minimum of G and can do no work, a cell that has reached metabolic equilibrium is dead!
The fact that metabolism as a whole is never at equilibrium is one of the defining features of life.

▼ **Figure 8.8 Equilibrium and work in open systems.**

(a) **An open hydroelectric system.** Water flowing through a turbine keeps driving the generator because intake and outflow of water keep the system from reaching equilibrium.



(b) **A multistep open hydroelectric system.** Cellular respiration is analogous to this system: Glucose is broken down in a series of exergonic reactions that power the work of the cell. The product of each reaction is used as the reactant for the next, so no reaction reaches equilibrium.

ATP AS AN ENERGY CARRIER

ATP is called a high-energy phosphate compound.

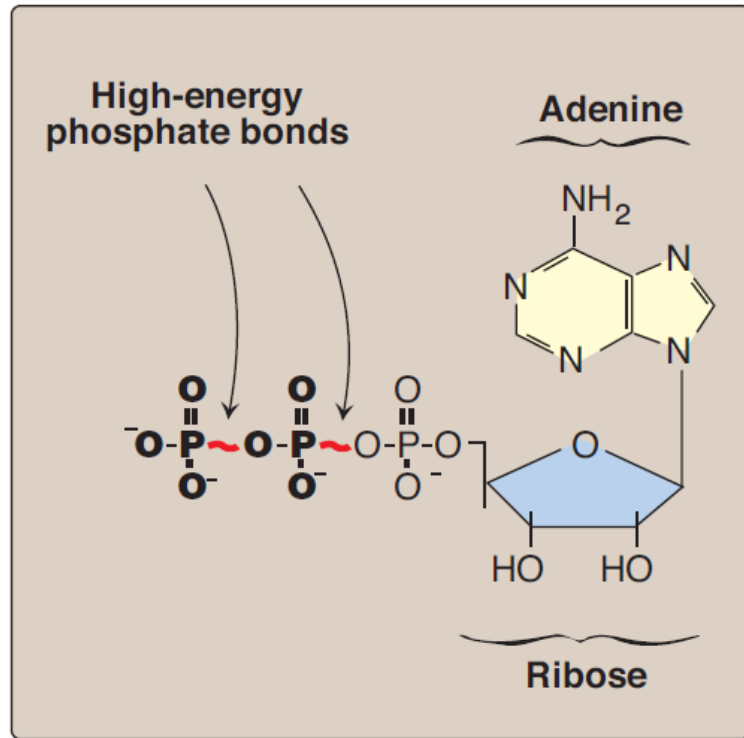
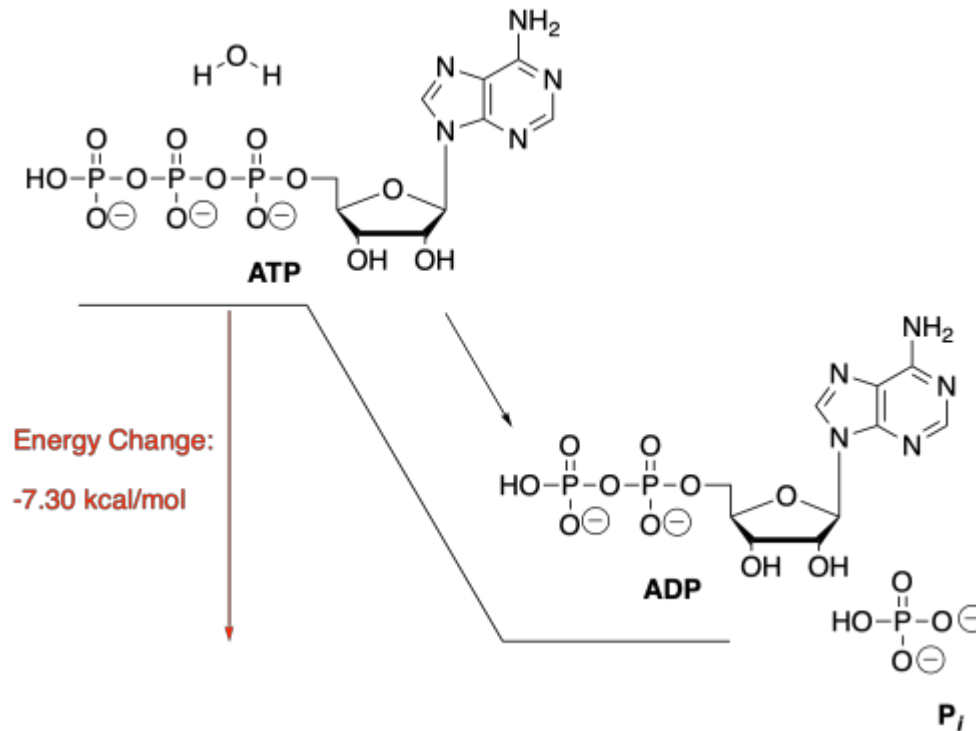
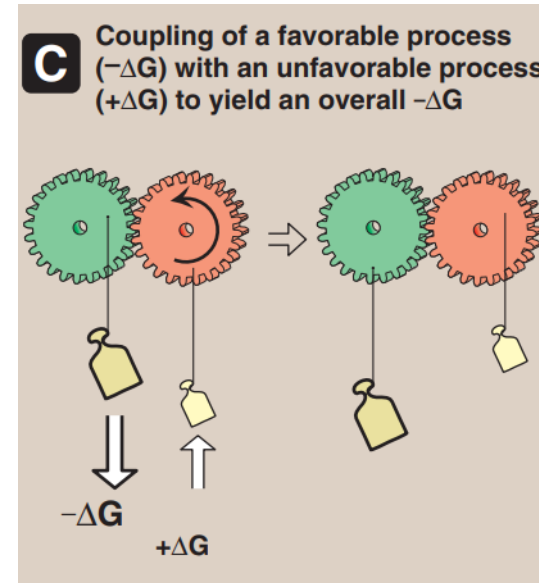
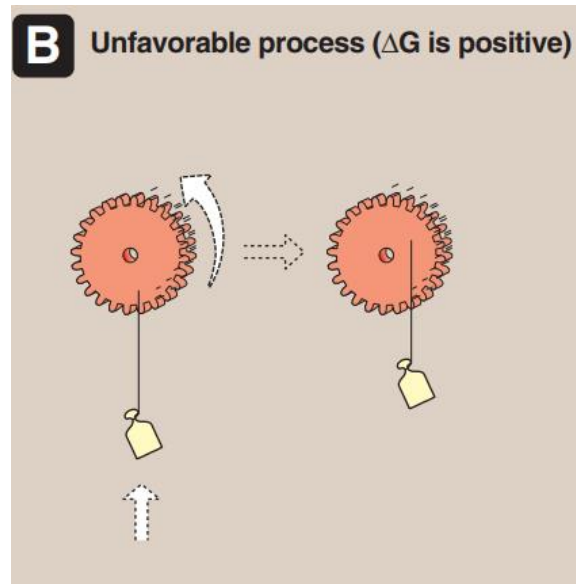
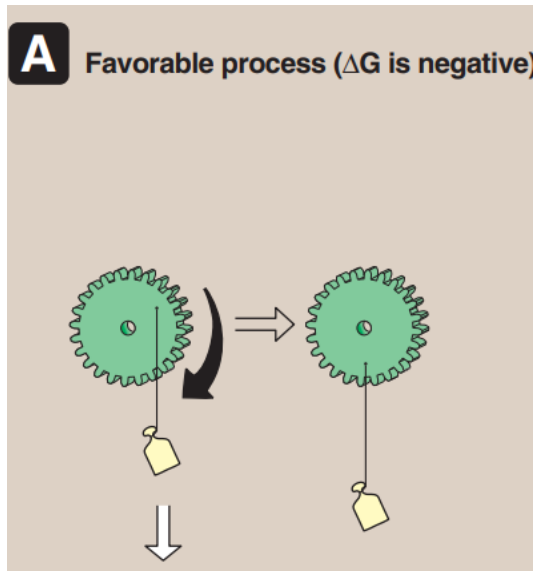
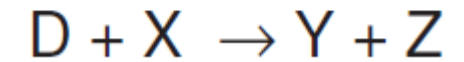
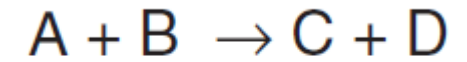


Figure 6.5
Adenosine triphosphate.



Reactions are coupled through common intermediates



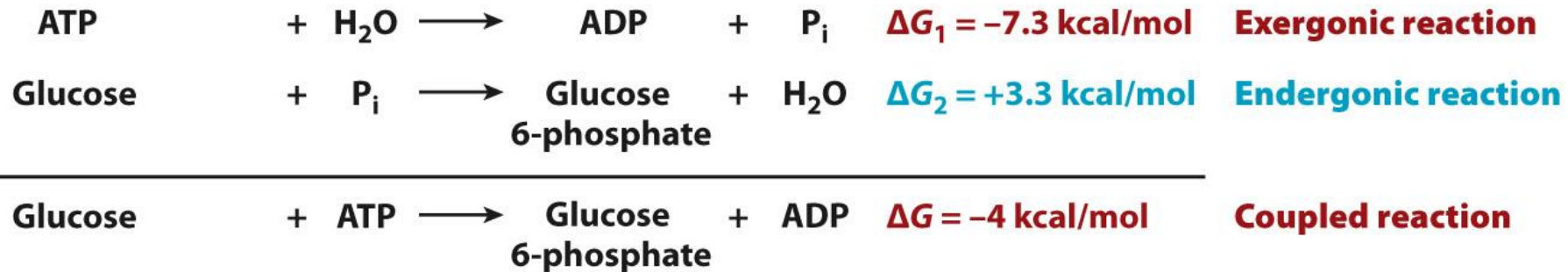
many coupled reactions use ATP to generate a common intermediate.

Energetic coupling

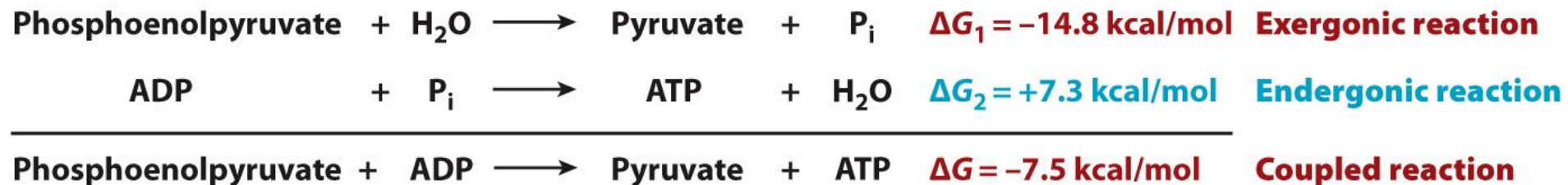
A spontaneous (exergonic) reaction drives a non-spontaneous (endergonic) reaction.

The coupled reaction proceeds because ΔG is negative and P_i is shared between the two reactions.

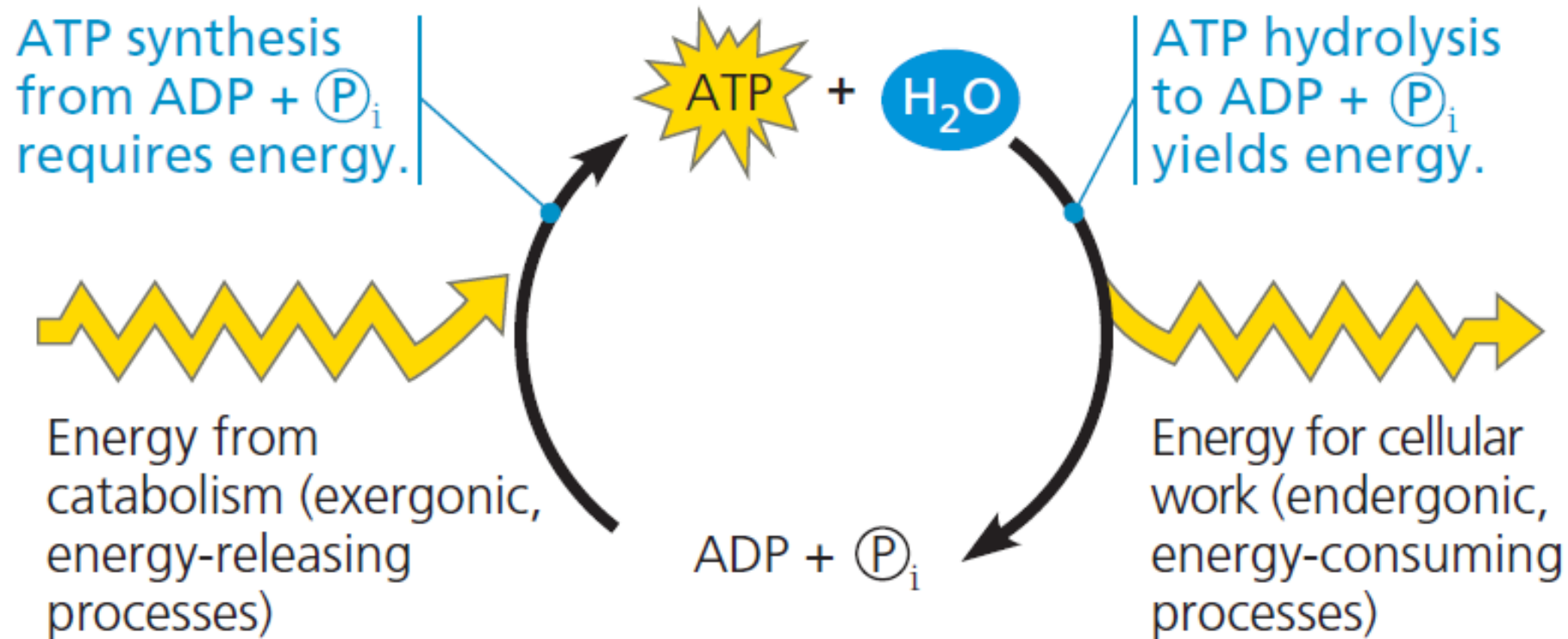
a.



b.



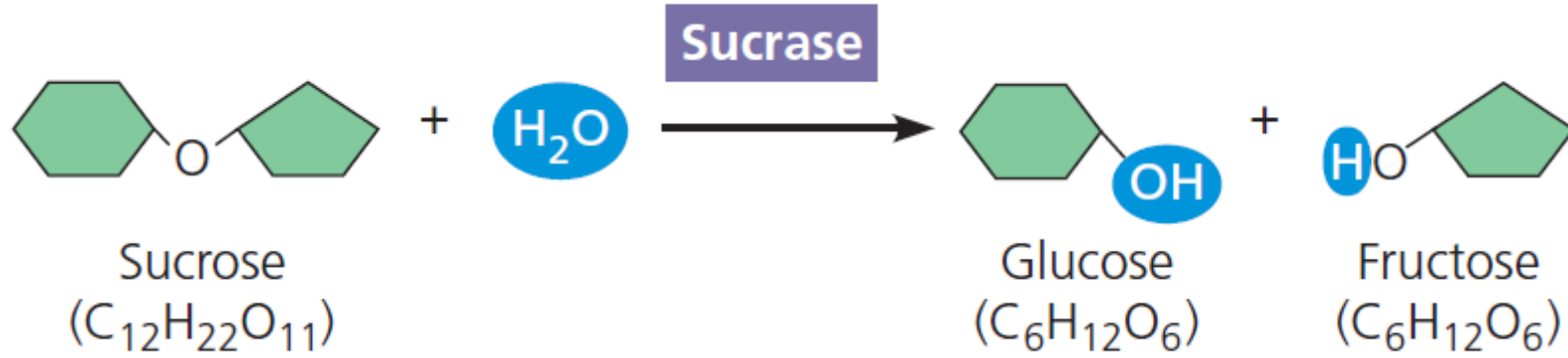
▼ **Figure 8.12 The ATP cycle.** Energy released by breakdown reactions (catabolism) in the cell is used to phosphorylate ADP, regenerating ATP. Chemical potential energy stored in ATP drives most cellular work.



Enzymes speed up metabolic reactions by lowering energy barriers

A spontaneous chemical reaction occurs without any requirement for outside energy, but it may occur so slowly that it is imperceptible.

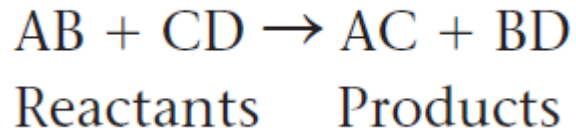
e.g. Hydrolysis of Sucrose is exergonic ($\Delta G = -7$ kcal/mol) but takes years. Adding sucrase enzyme can complete the reaction in seconds.



How does the enzyme do this?

Enzymes

- macromolecule that acts as a **catalyst**, a chemical agent that speeds up a reaction without being consumed by the reaction.
- They do not add any energy to reactions (hence cannot make an endergonic reaction work!)

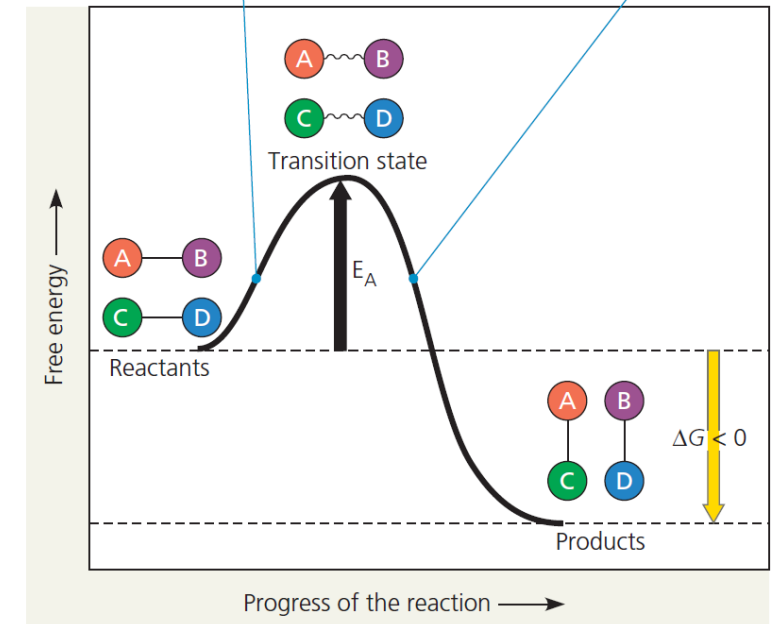


▼ **Figure 8.13 Energy profile of an exergonic reaction.**

The “molecules” are hypothetical, with A, B, C, and D representing portions of the molecules. Thermodynamically, this is an exergonic reaction, with a negative ΔG , and the reaction occurs spontaneously. However, the activation energy (E_A) provides a barrier that determines the rate of the reaction.

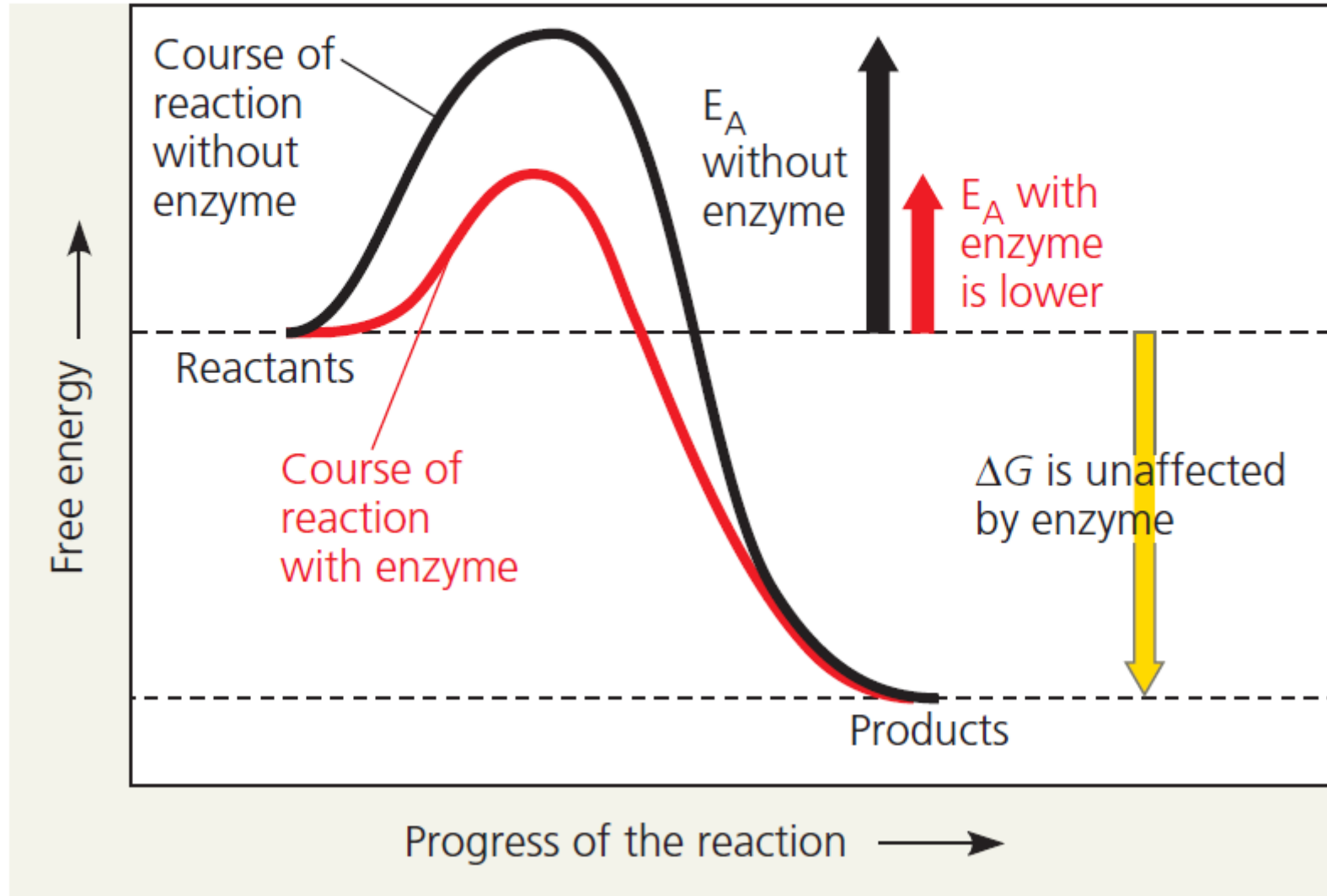
The reactants AB and CD must absorb enough energy from the surroundings to reach the unstable transition state, where bonds can break.

After bonds have broken, new bonds form, releasing energy to the surroundings.



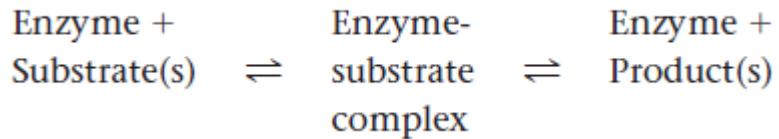
▼ **Figure 8.14 The effect of an enzyme on activation energy.**

Without affecting the free-energy change (ΔG) for a reaction, an enzyme speeds the reaction by reducing its activation energy (E_A).



an enzyme cannot change the G for a reaction; it cannot make an endergonic reaction exergonic would eventually occur anyway, but this enables the cell to have a dynamic metabolism, routing chemicals smoothly through metabolic pathways. Also, enzymes are very specific for the reactions they catalyze, so they determine which chemical processes will be going on in the cell at any given time.

Substrate Specificity of Enzymes

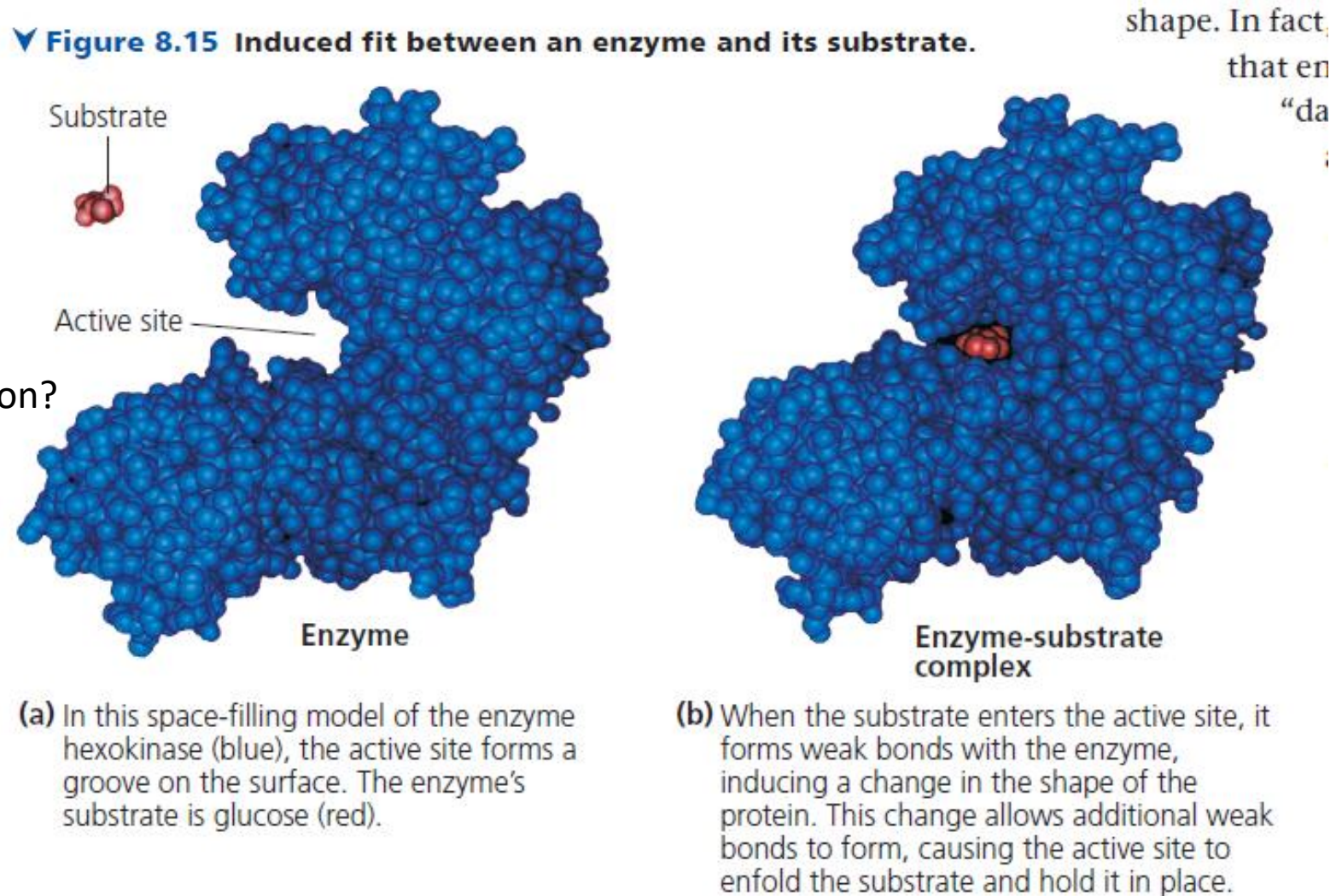


each enzyme is very specific

What accounts for this molecular recognition?

The specificity of an enzyme is attributed to a complementary fit between the shape of its active site and the shape of the substrate.

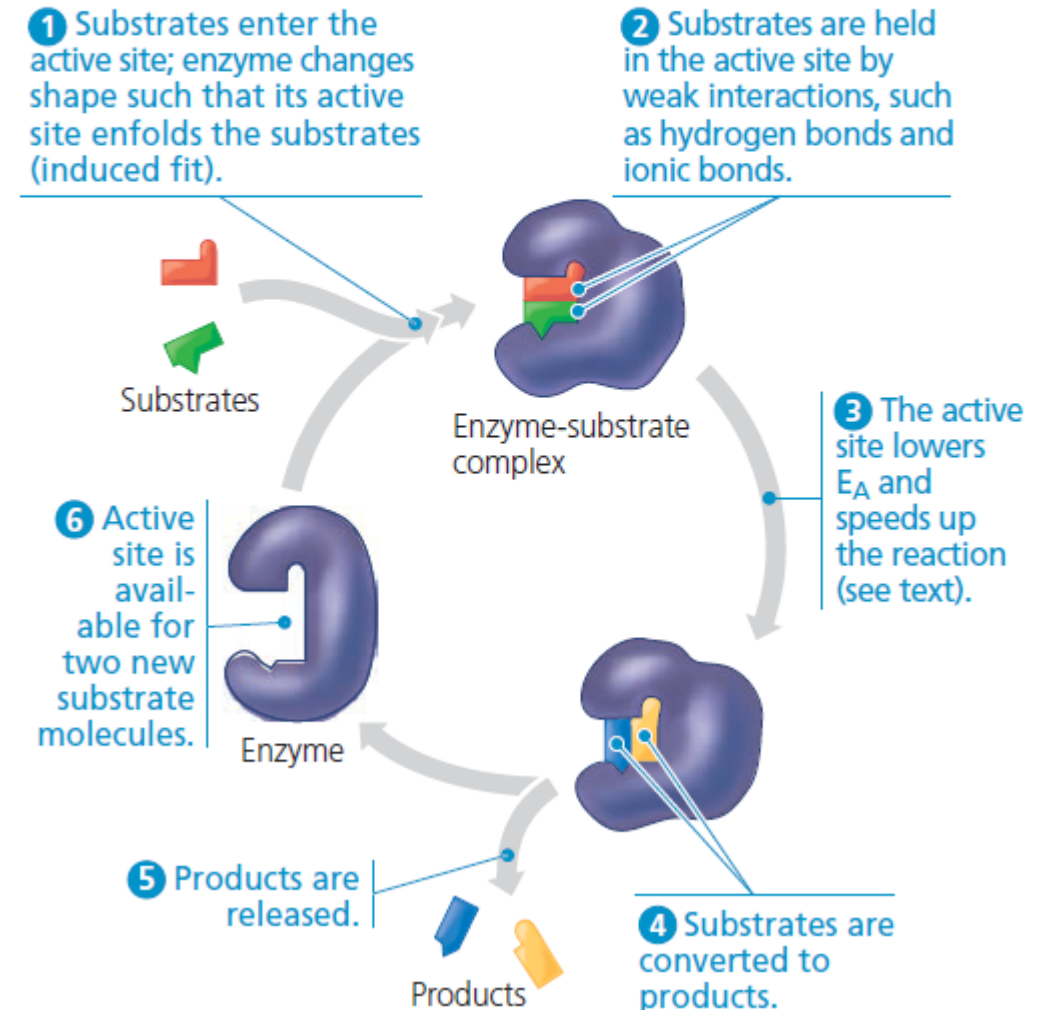
- The induced-fit model is an adjustment to the lock-and-key model and explains how enzymes and substrates undergo dynamic modifications during the transition state to increase the affinity of the substrate for the active site.



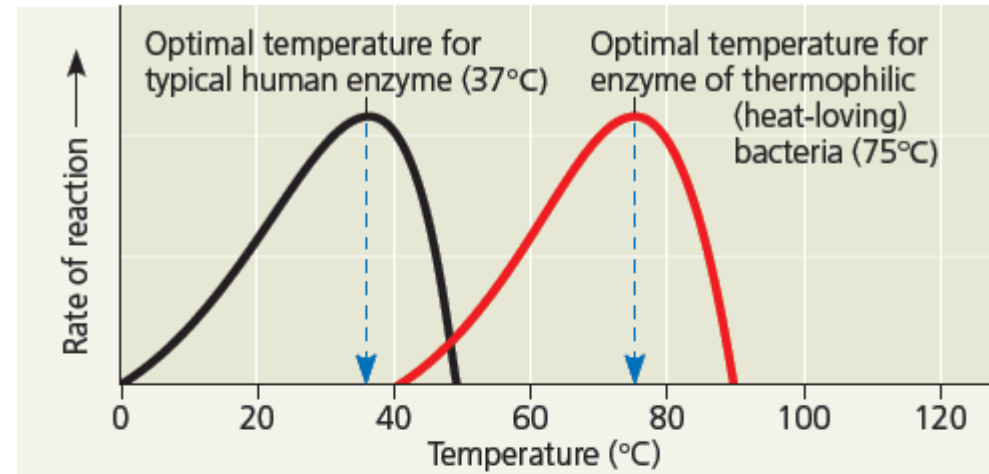
Catalysis in the Enzyme's Active Site

▼ Figure 8.16 The active site and catalytic cycle of an enzyme.

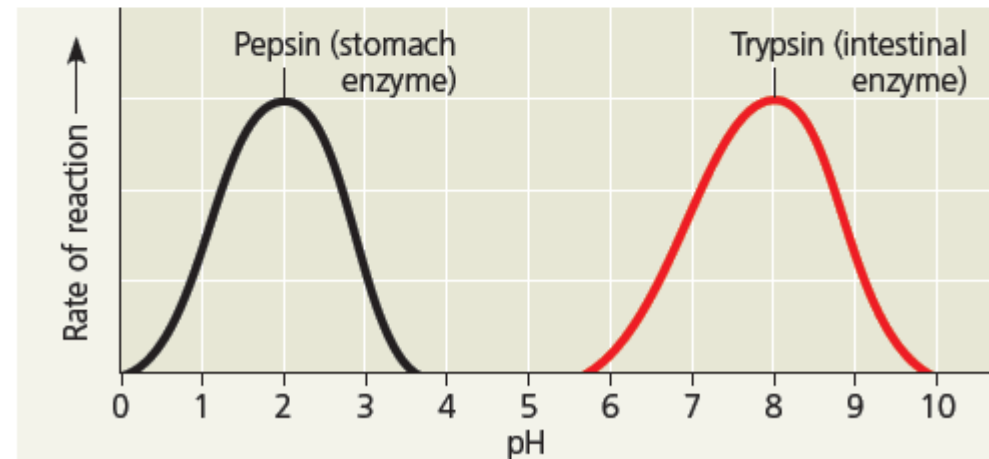
An enzyme can convert one or more reactant molecules to one or more product molecules. The enzyme shown here converts two substrate molecules to two product molecules.



Factors affecting enzyme activity



(a) The photo shows thermophilic cyanobacteria (green) thriving in the hot water of a Nevada geyser. The graph compares the optimal temperatures for an enzyme from the thermophilic bacterium *Thermus oshimai* (75°C) and human enzymes (body temperature, 37°C).



(b) This graph shows the rate of reaction for two digestive enzymes over a range of pH values.

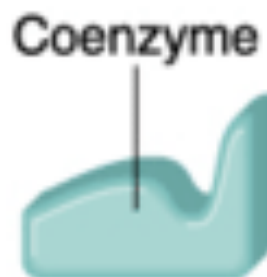
Enzyme regulation

- Or controlling a chemical reaction through changing the enzyme activity (!)
 - Enzyme activation – by favoring the binding of substrate to the active site
 - Enzyme inactivation (inhibition) – by blocking the substrate from binding to active site
- Help treating diseases by developing drugs that can change the enzyme activity
 - Very efficient since substrates and enzymes are HIGHLY specific

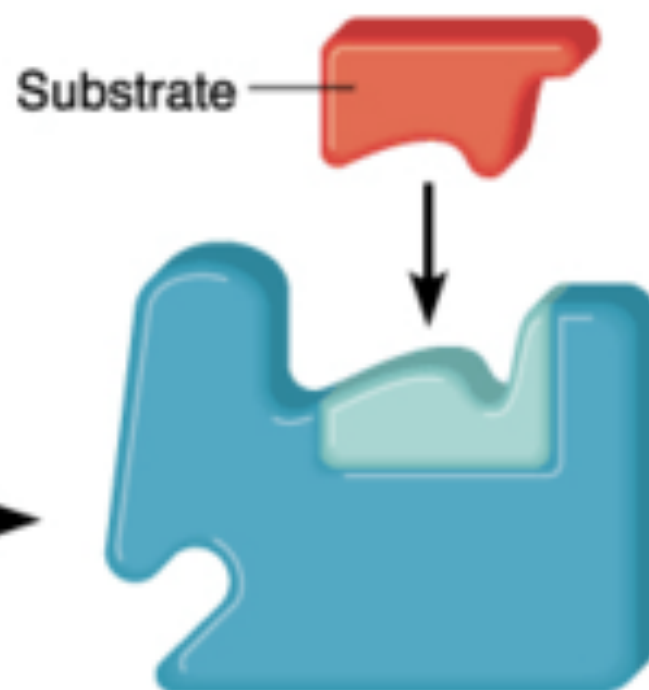


Apoenzyme
(protein portion),
inactive

+



Cofactor
(nonprotein portion),
activator

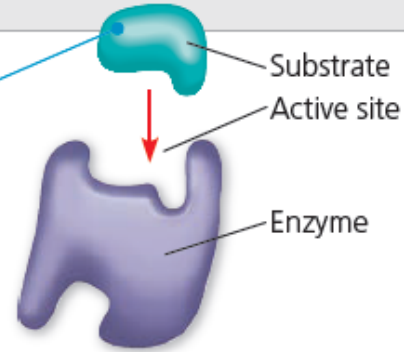


Holoenzyme
(whole enzyme),
active

▼ **Figure 8.18** Inhibition of enzyme activity.

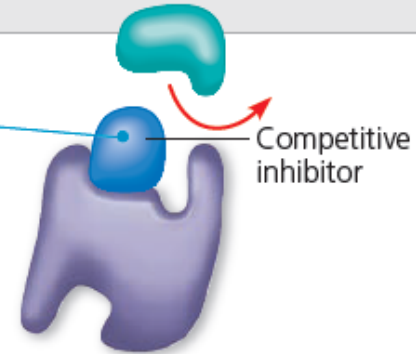
(a) Normal binding

A substrate can bind normally to the active site of an enzyme.



(b) Competitive inhibition

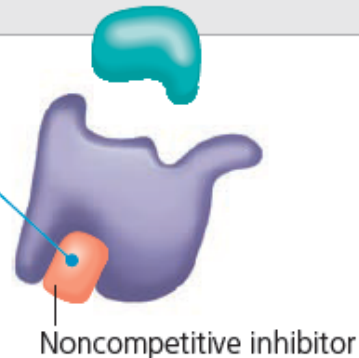
A competitive inhibitor mimics the substrate, competing for the active site.



 **Animation: Enzymes: Competitive Inhibition**

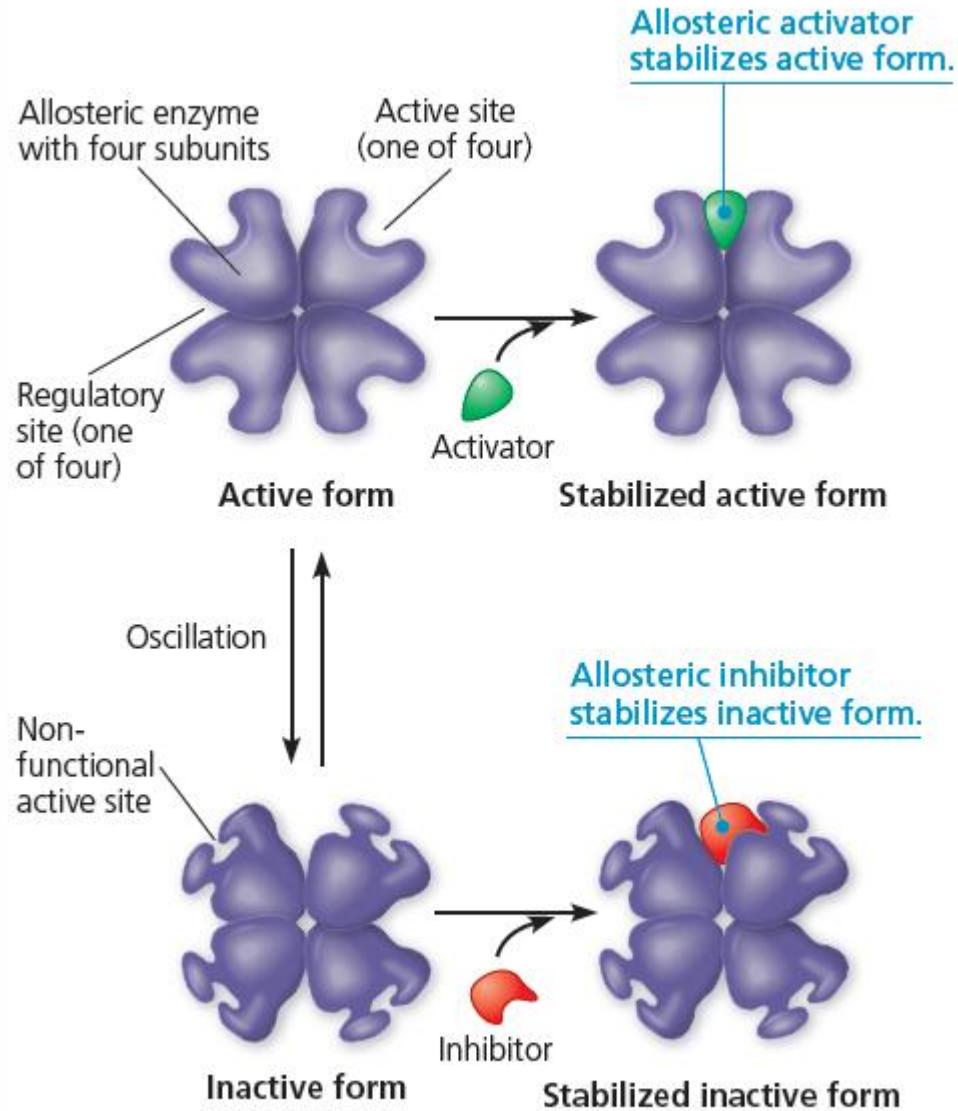
(c) Noncompetitive inhibition

A noncompetitive inhibitor binds to the enzyme away from the active site, altering the shape of the enzyme so that even if the substrate can bind, the active site functions less effectively, if at all.



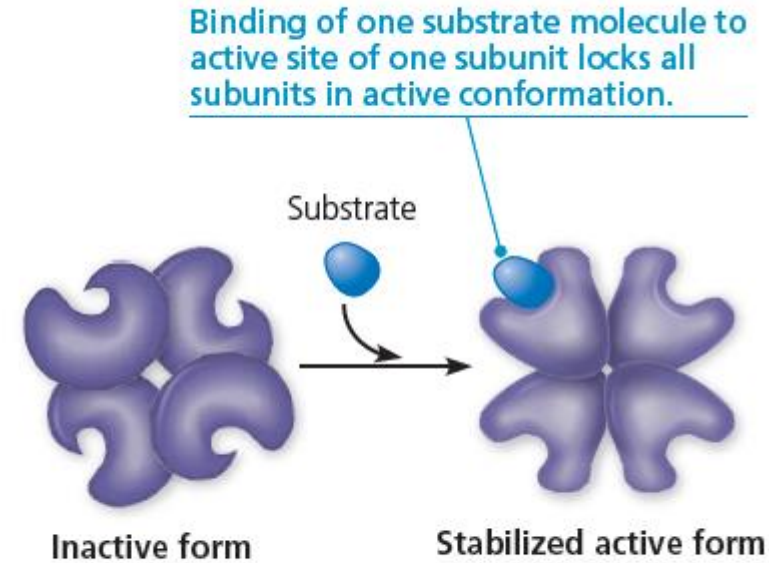
 **Animation: Enzymes: Noncompetitive Inhibition**

(a) Allosteric activators and inhibitors



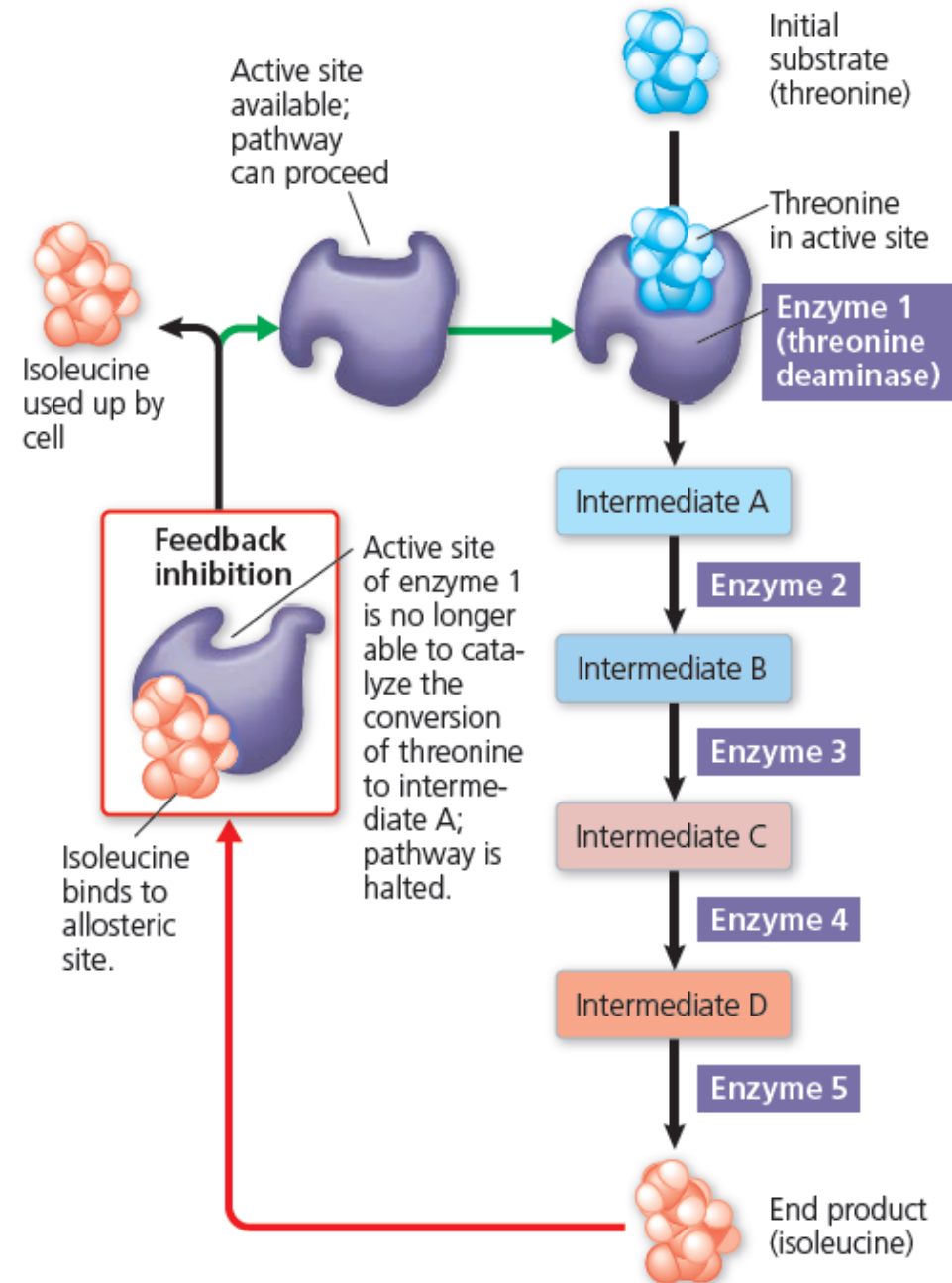
At low concentrations, activators and inhibitors dissociate from the enzyme. The enzyme can then oscillate again.

(b) Cooperativity: another type of allosteric activation

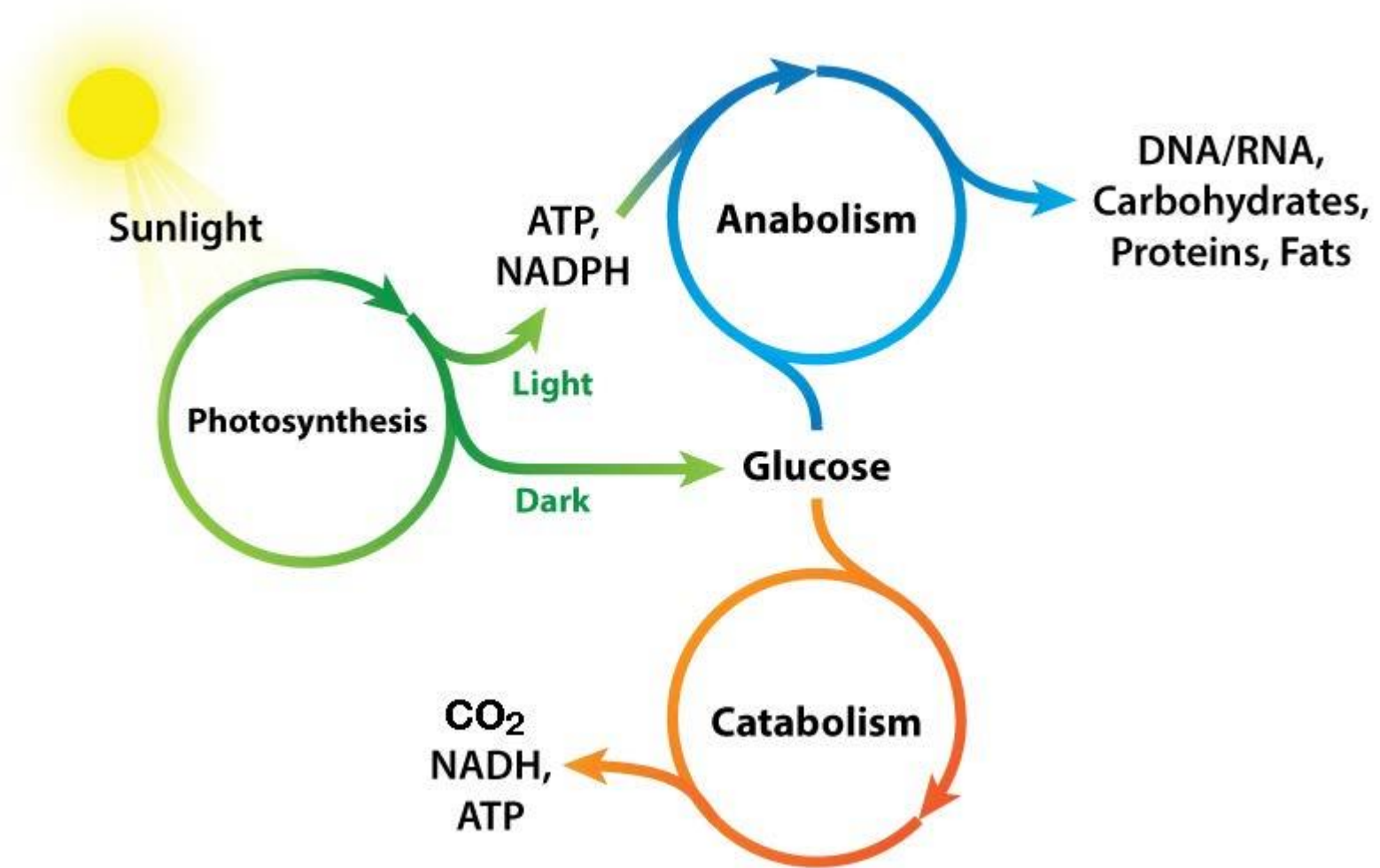


The inactive form shown on the left oscillates with the active form when the active form is not stabilized by substrate.

▼ **Figure 8.21** Feedback inhibition in isoleucine synthesis.



Energy flow and chemical recycling in ecosystems.

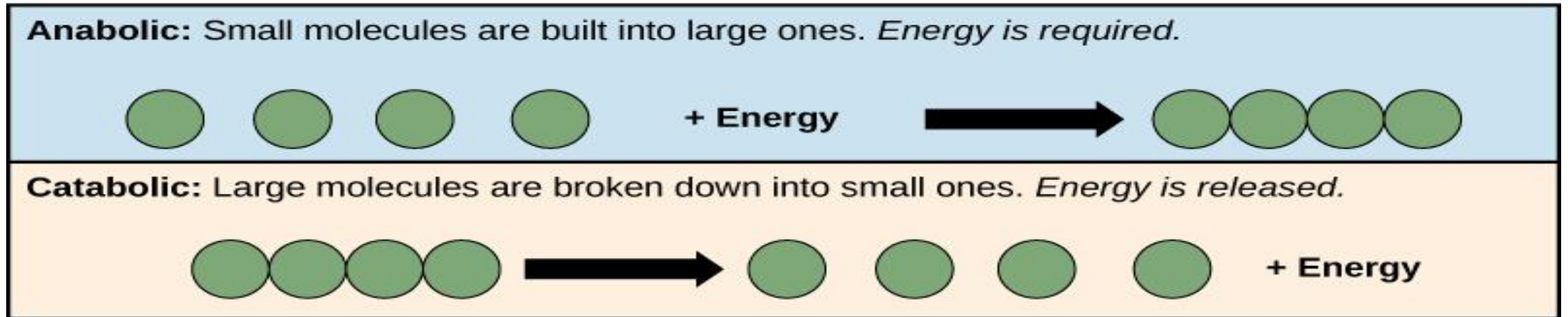


Metabolism

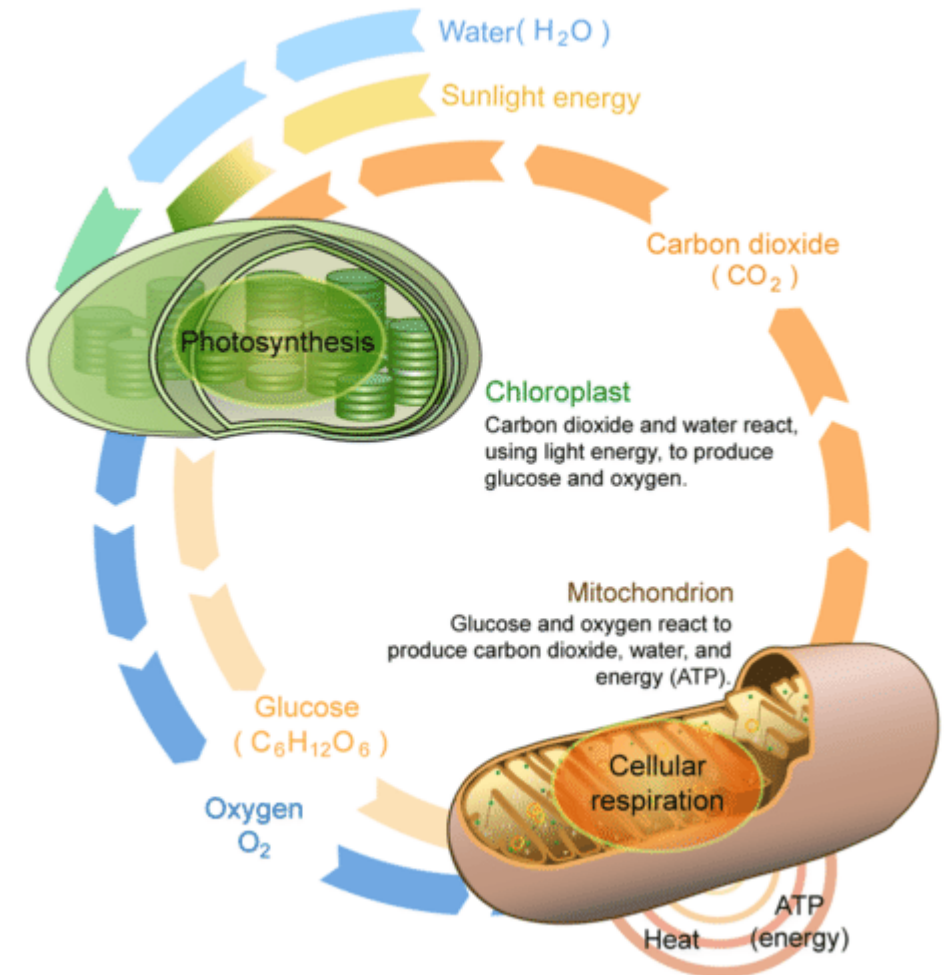
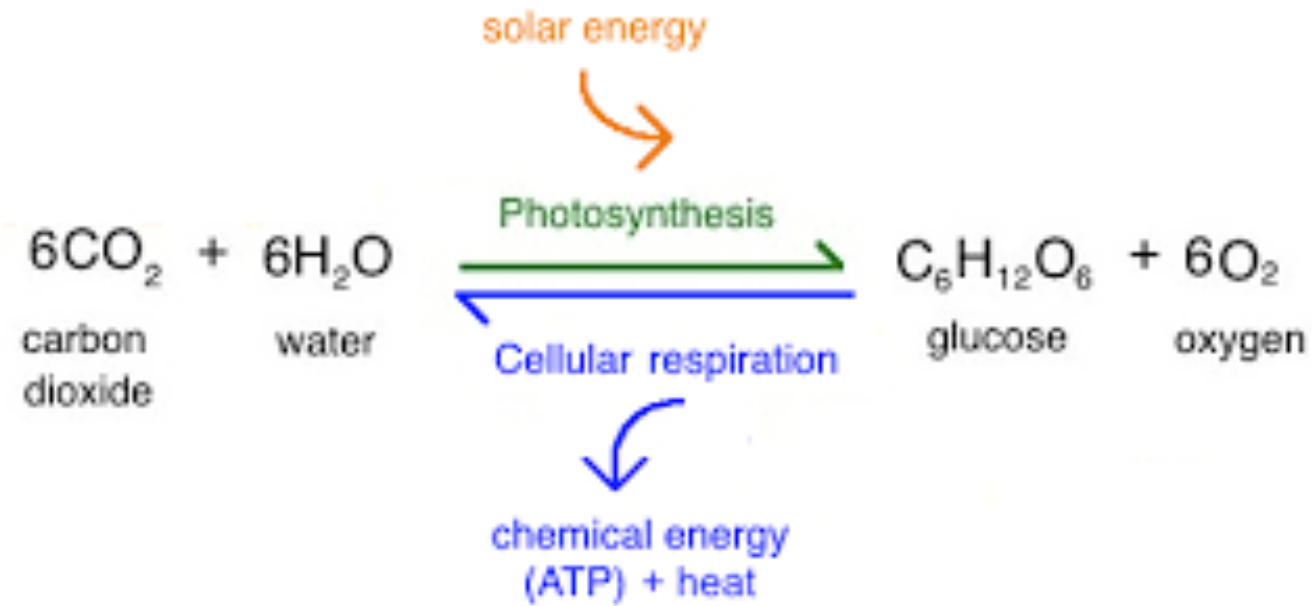
- Metabolism = represents all the chemical reactions and processes taking place inside your cells resulting in LIFE.
- Catabolism = chemical reactions resulting in breaking down big molecules
 - releases energy
- Anabolism = chemical reactions resulting in building up molecules
 - requires energy
- Metabolic pathway = all the consecutive reactions resulting in a specific product
 - Examples:.....

Anabolism vs. Catabolism

Metabolic pathways

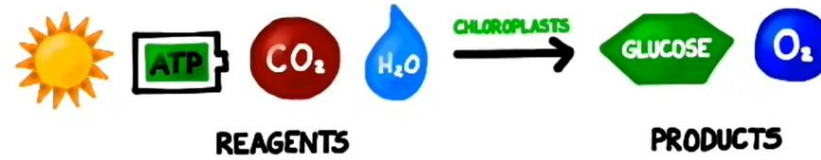


- Catabolic pathways are those that generate energy by breaking down larger molecules. Anabolic pathways are those that require energy to synthesize larger molecules. Both types of pathways are required for maintaining the cell's energy balance.

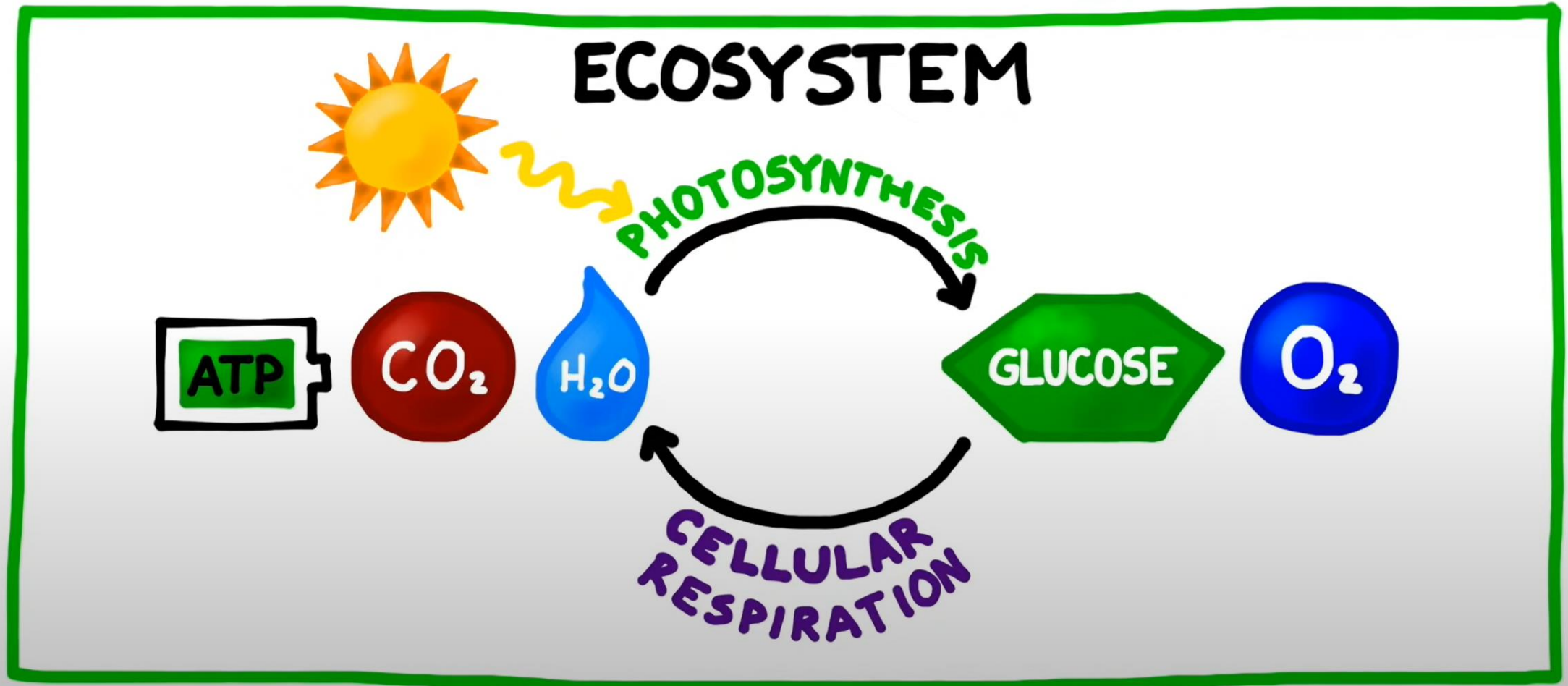


PHOTOSYNTHESIS

REAGENTS +
PRODUCTS:



CELLULAR RESPIRATION



PHOTOSYNTHESIS

LOCATION:

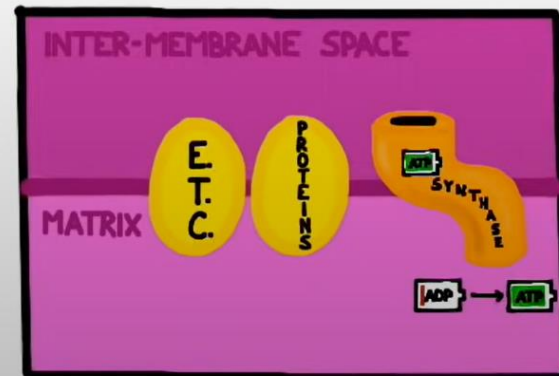
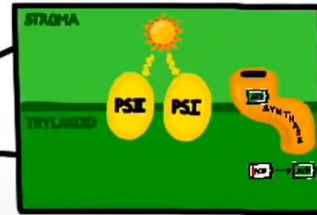
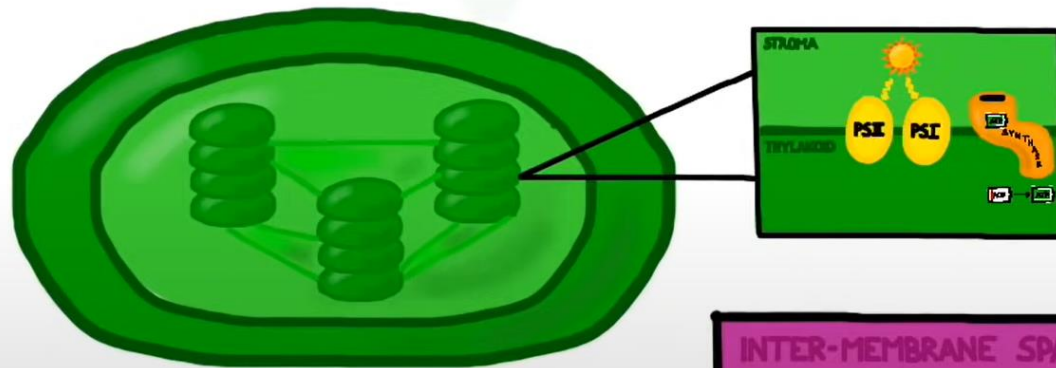
CHLOROPLASTS

INNER
COMPARTMENT:

THYLAKOID

OUTER
COMPARTMENT:

STROMA



CELLULAR RESPIRATION

CYTOPLASM + MITOCHONDRIA

MATRIX

INTER-MEMBRANE SPACE

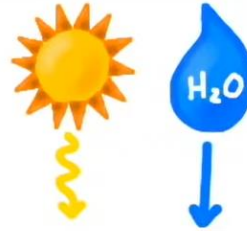


[Bing Videos:](#)
Photosynthesis

MAJOR
STEPS:

PHOTOSYNTHESIS

PHOTO:

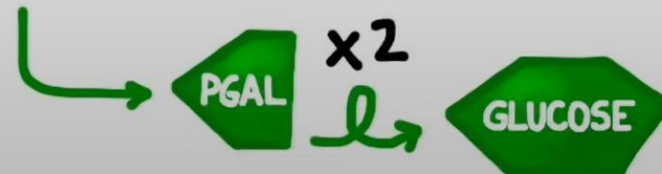


LIGHT DEPENDENT
REACTIONS

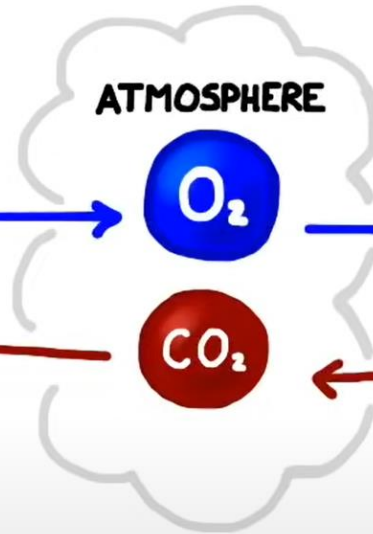
SYNTHESIS:



LIGHT
INDEPENDENT
REACTIONS



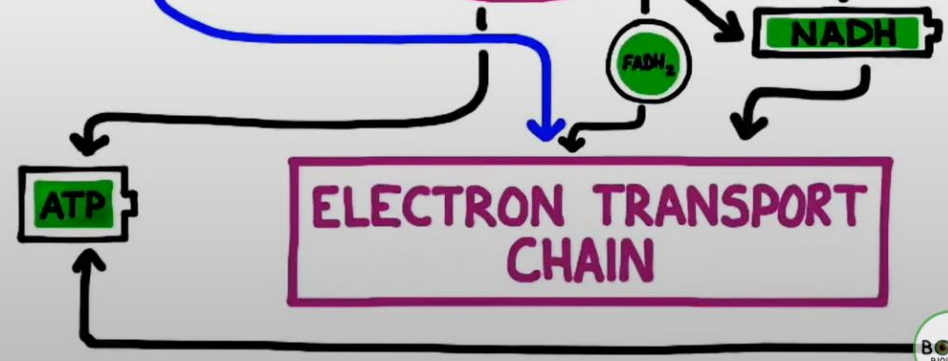
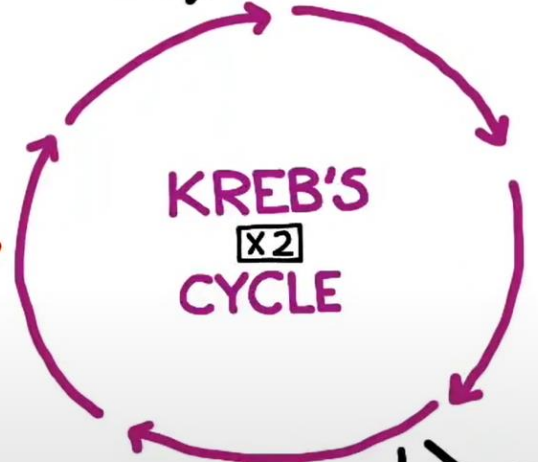
[Bing Videos:](#)
Calvin Cycle



CELLULAR RESPIRATION



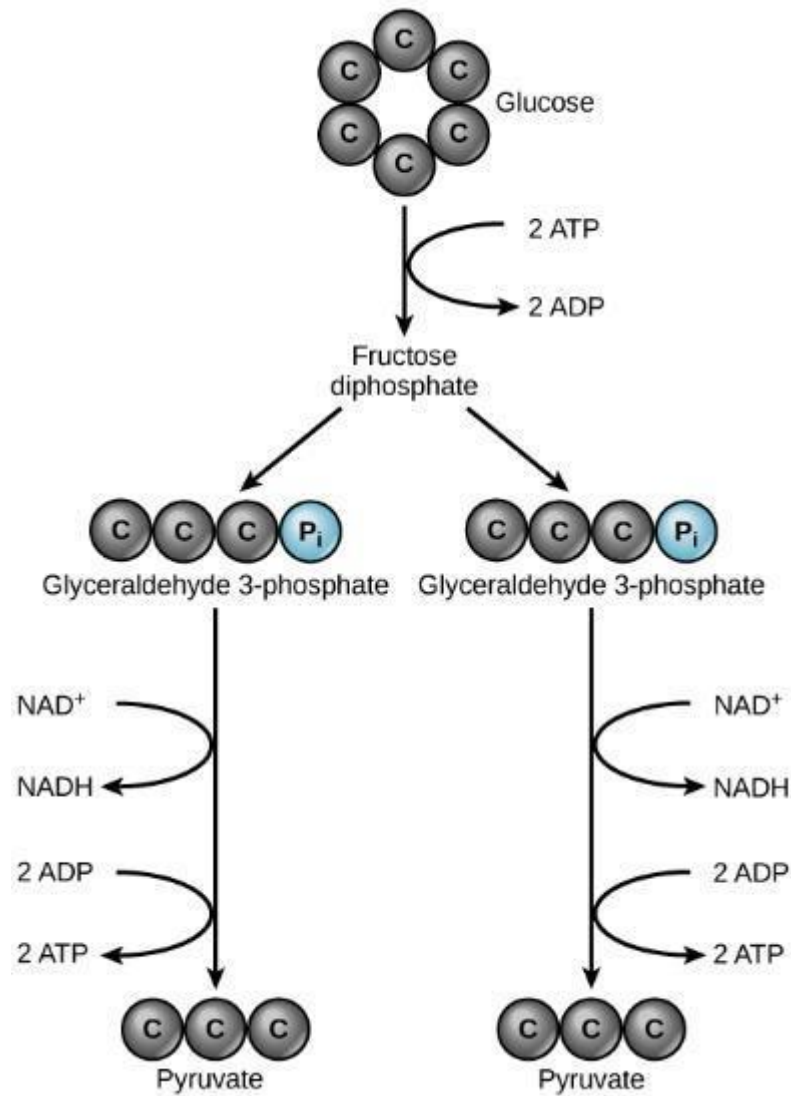
GLYCOLYSIS



Cellular respiration

- Steps:
 1. Glycolysis
 2. Pyruvate processing
 3. Citric cycle
 4. Oxidative phosphorylation

glycolysis



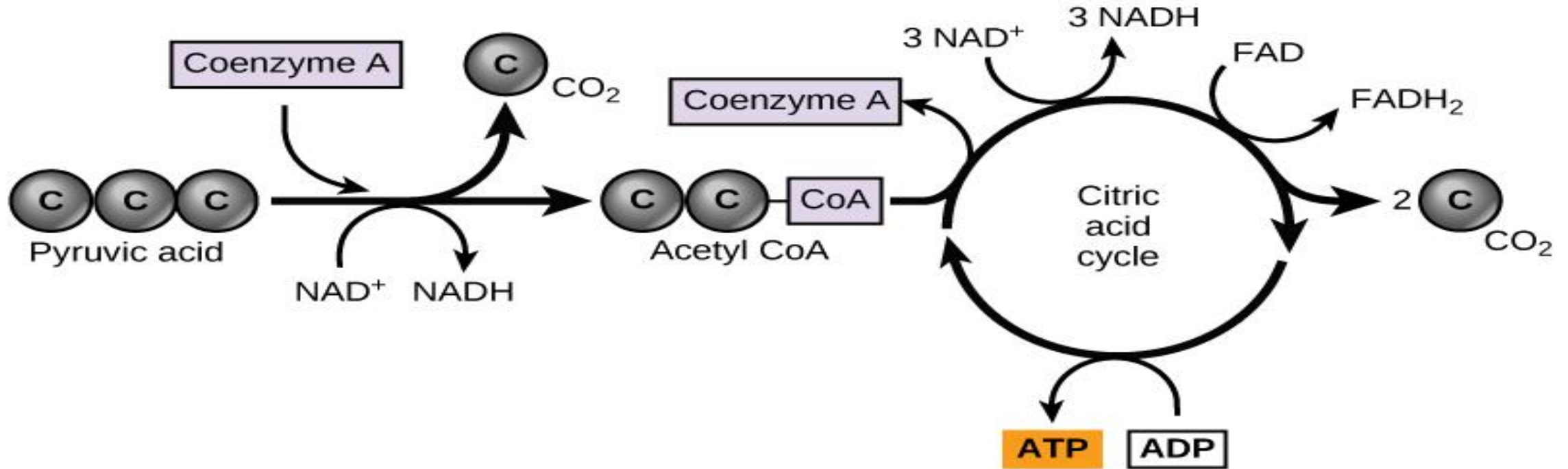
- In glycolysis, a glucose molecule is converted into two pyruvate molecules.
- 6 carbons-molecule (glucose) is transformed into 2 pyruvates (3-carbon-molecules)
- 1st phase – requires energy input
- 2nd phase – makes energy (ATP, NADH)

[Bing Videos](#)

Pyruvate processing & citric cycle

- EACH Pyruvate molecule is processed:
 - 1 carbon is lost as CO₂
 - 2 carbons are transformed into Acetyl-Coenzyme A
 - Makes energy as NADH
- Citric cycle:
 - Each carbon from Acetyl-CoA is lost as CO₂
 - Makes energy as ATP, NADH, FADH₂

pyruvate & Citric Cycle



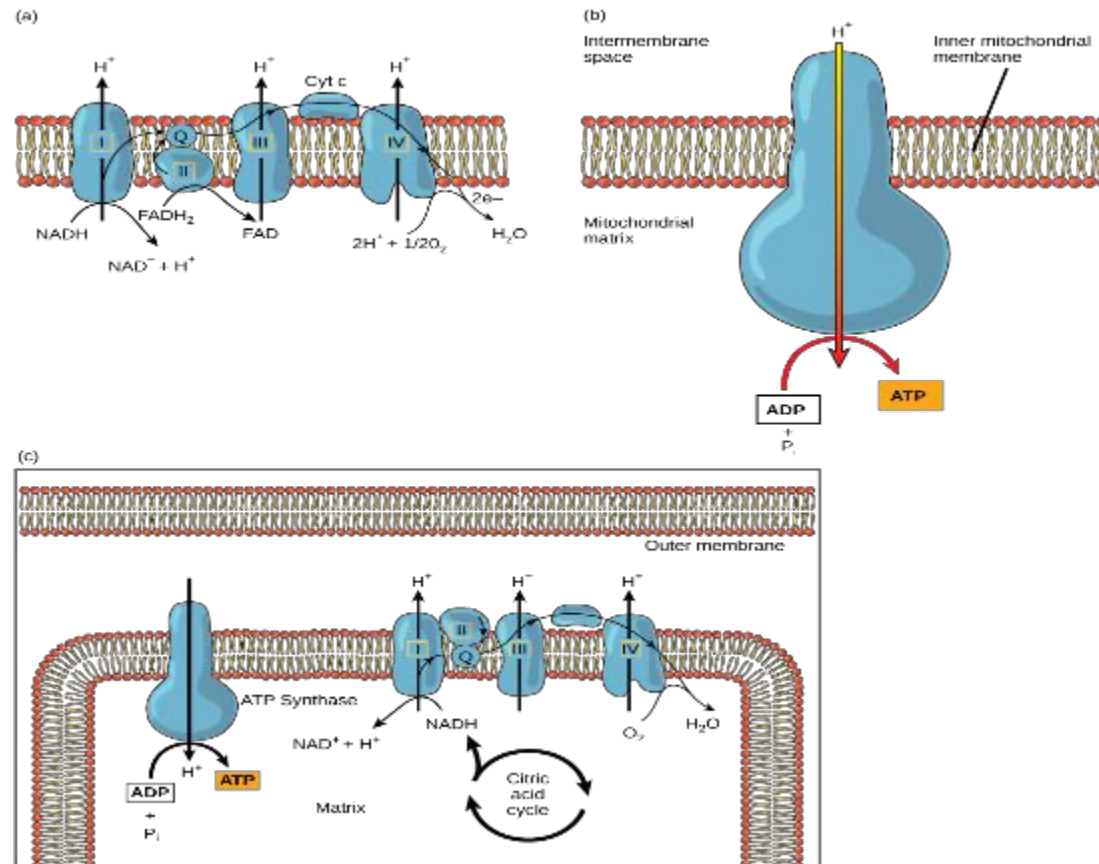
- Pyruvate is converted into acetyl-CoA before entering the citric acid cycle.

[Bing Videos](#)

Glycolysis, Pyruvate processing & citric cycle

- Overall:
 - Follow the carbon: 6-carbon atoms in 1 glucose molecule becomes 6 CO₂ molecules (exhaled out during respiration)
- Energy-containing molecules are made:
 - ATP
 - NADH, FADH₂
 - These will provide energy for the next step (!)

oxidative phosphorylation

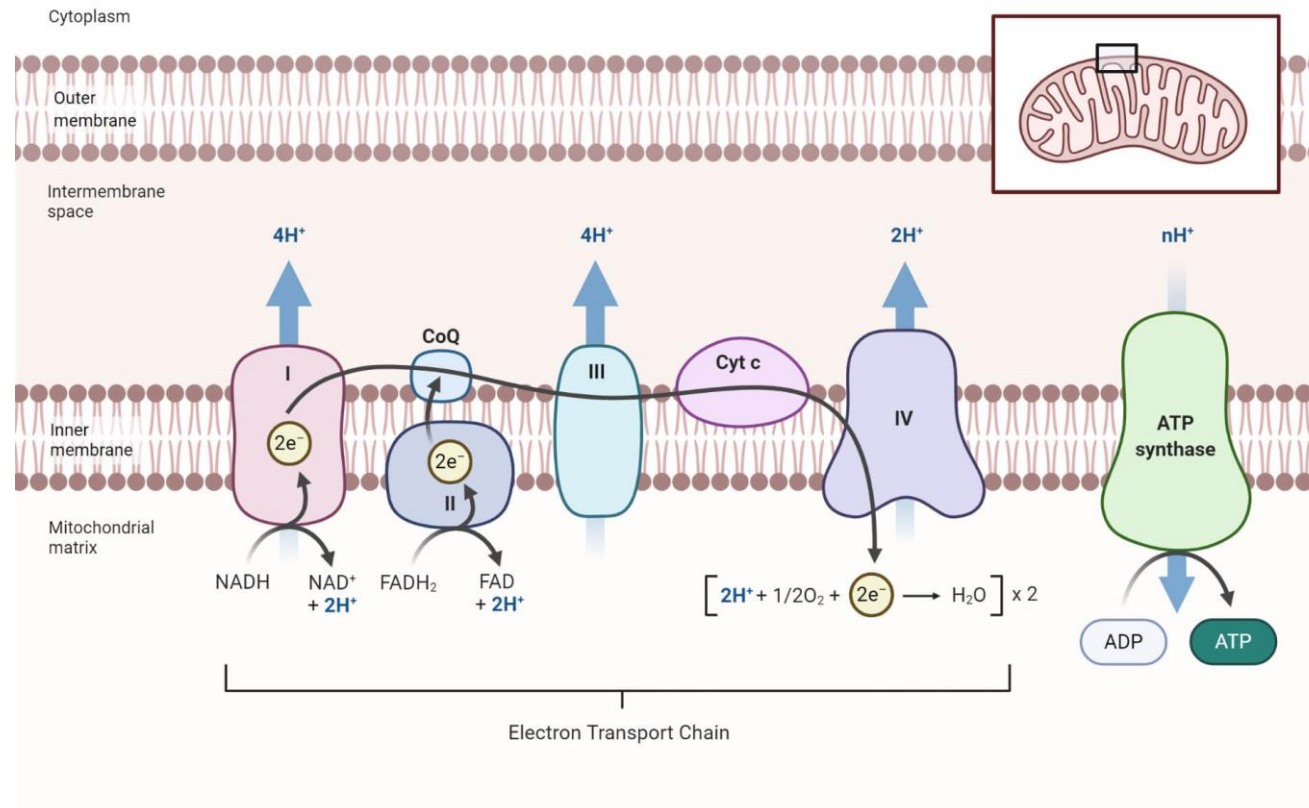
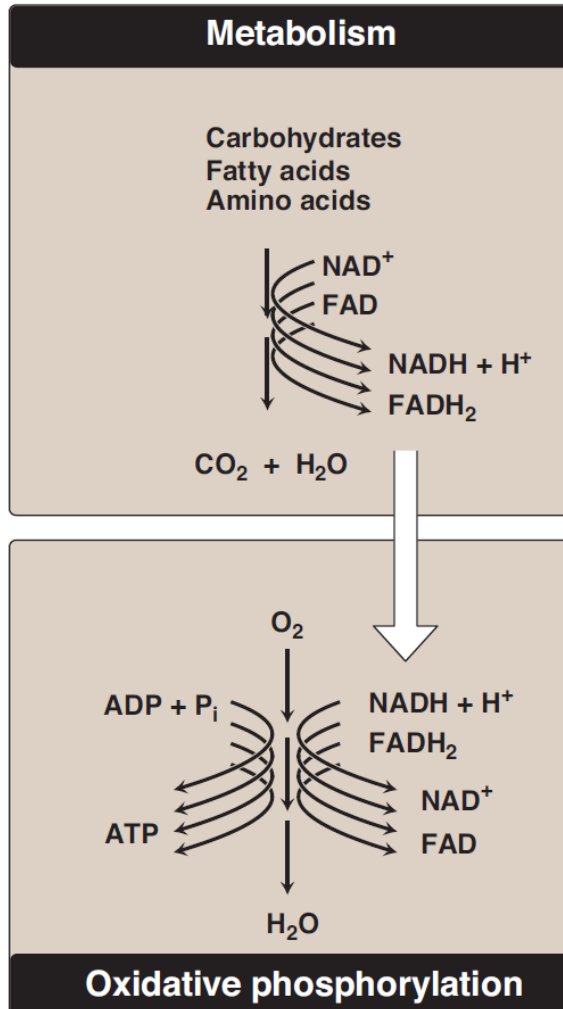


- (a) The electron transport chain is a set of molecules that supports a series of oxidation-reduction reactions.
- (b) ATP synthase is a complex, molecular machine that uses an H^+ gradient to regenerate ATP from ADP.
- (c) Chemiosmosis relies on the potential energy provided by the H^+ gradient across the membrane.

Oxidative phosphorylation

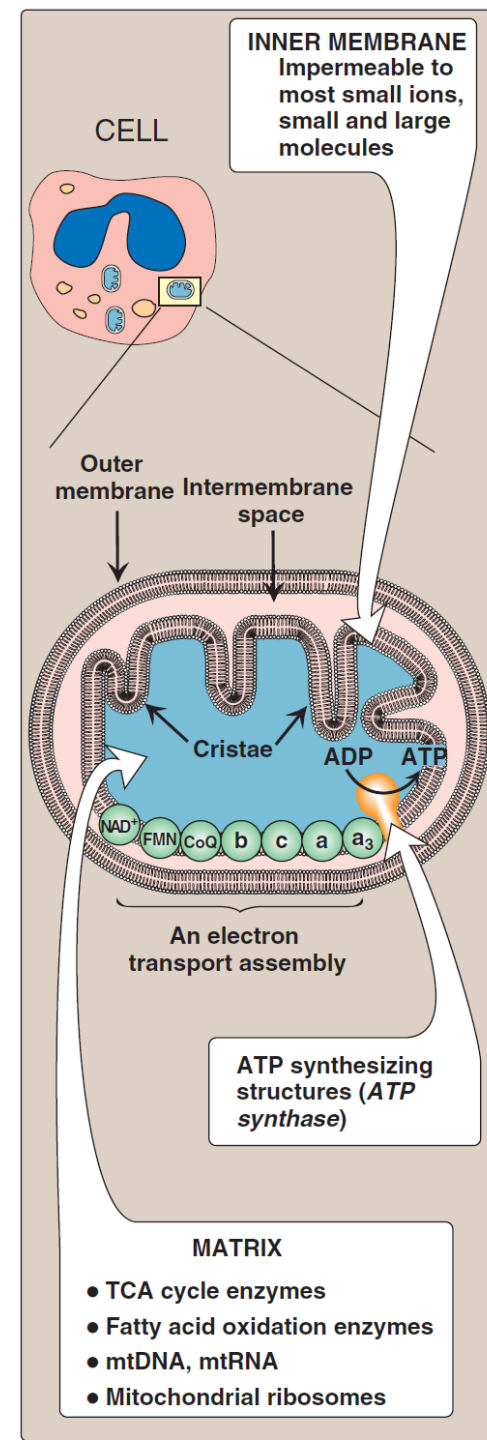
- Oxidation:
 - Losing electrons
 - NADH & FADH₂ will donate (this is oxidation) electrons to the ETC
 - ETC = electron transport chain
 - Made of a chain of proteins
 - Will use the energy of the electrons to create a H⁺ gradient
 - Moves H⁺ across the membrane resulting in higher concentration in the intermembrane space than the matrix
 - Will transport the electrons to O₂ making H₂O
- Phosphorylation:
 - Process of adding phosphate to a molecule
 - H⁺ will passively move down the gradient through ATP-synthase (from intermembrane space back to matrix)
 - This makes ATP-synthase create ATP from ADP + 1 P
 - 90% of ATP (energy) is made this way
 - Chemiosmosis = coupling of chemical reaction of making ATP to passive transport of H⁺ down their concentration gradient

ELECTRON TRANSPORT CHAIN



[Bing Videos: ETC](#)

[Bing Videos: ATP synthase](#)



Fermentation & energy

- Fermentation – obtaining energy without the need of O₂
 - Makes less ATP than cellular respiration
 - Sugar goes only through glycolysis (!)
- Two types:
 - Lactic acid fermentation – in muscles
 - Alcohol fermentation – in yeast – makes alcohol and gas

- Lactic acid fermentation is common in muscles that have become exhausted by use.

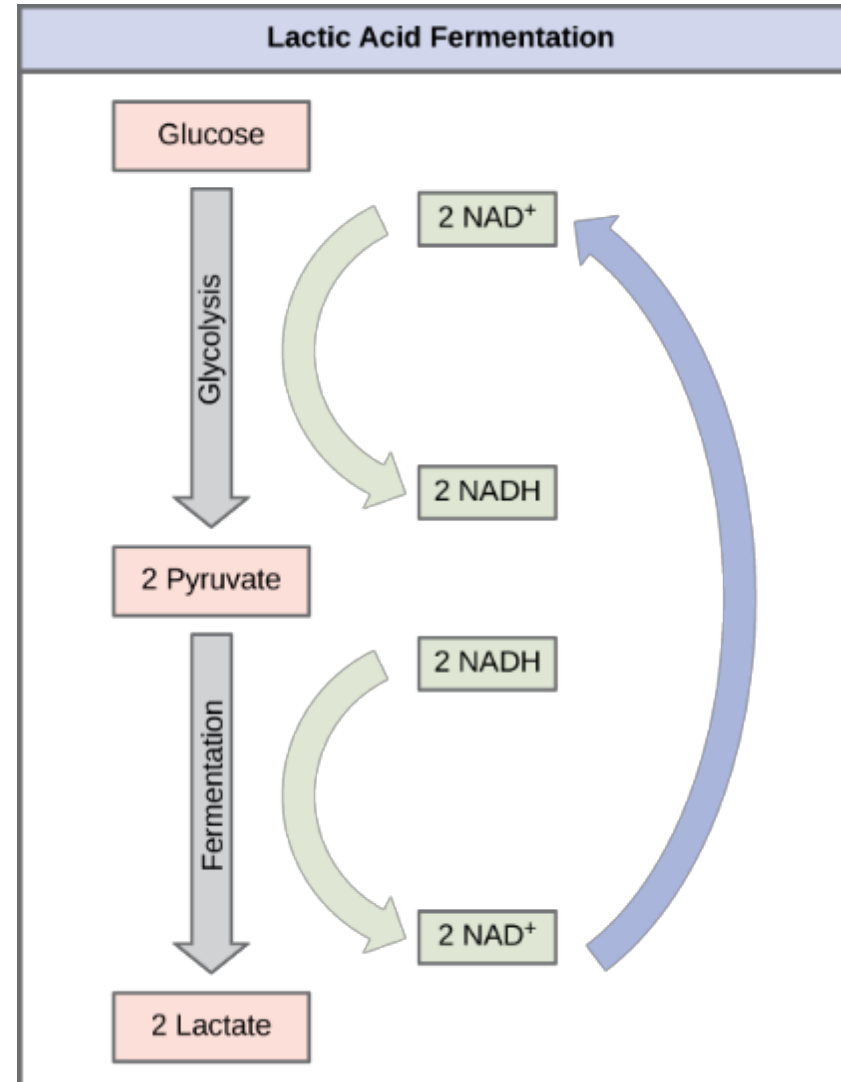
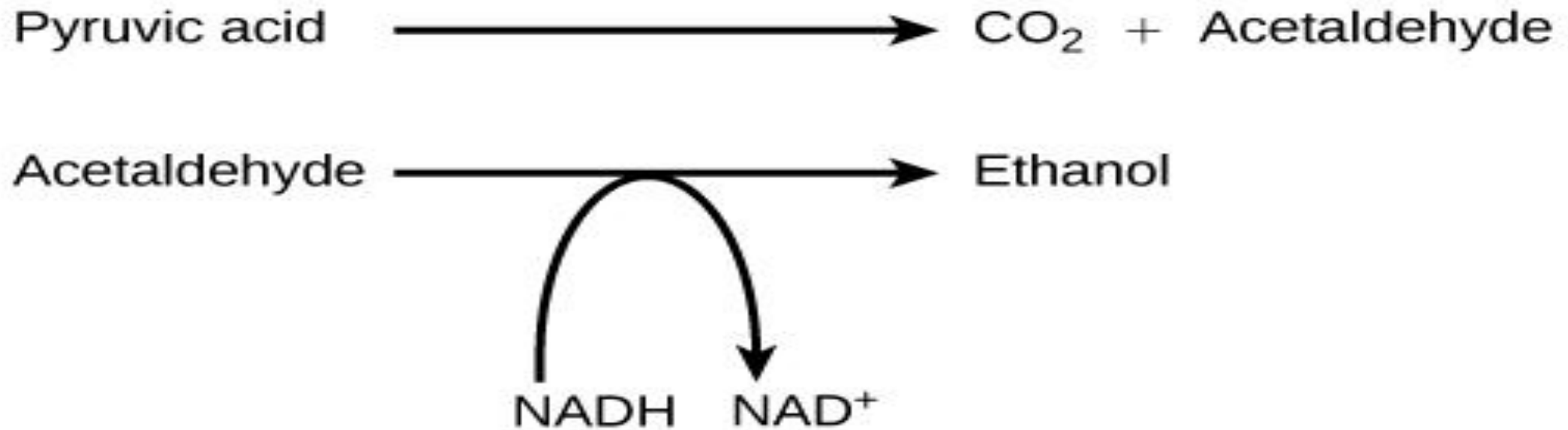


Figure 4.17



- The reaction resulting in alcohol fermentation is shown.



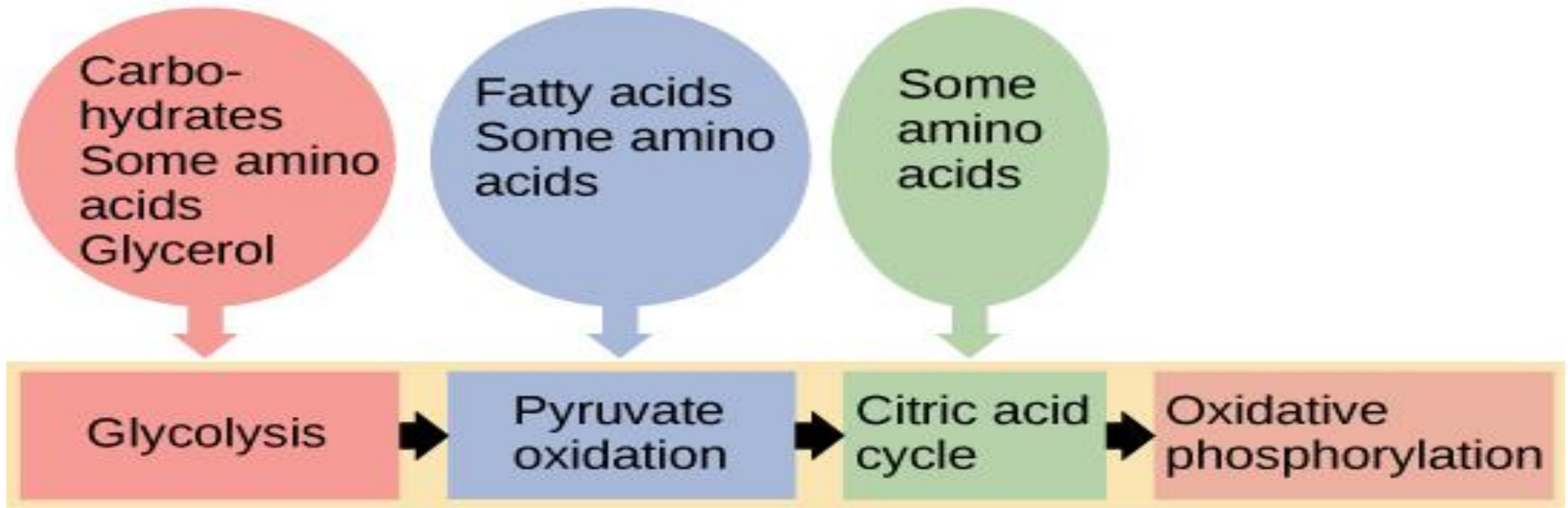
- Fermentation of grape juice to make wine produces CO_2 as a byproduct. Fermentation tanks have valves so that pressure inside the tanks can be released.

Anaerobic respiration

- Making energy without the use of O₂
- Uses inorganic molecules like:
 - Sulfates to make sulfide gas (H₂S) – bacteria in water wells
 - CO₂ to make methane gas (CH₄) – stomach ruminants (cows)
- Present in bacteria & archaea

Macromolecules & energy

- How does a turkey sandwich, which contains protein, provide energy to your cells?
 - This happens because all the catabolic pathways for carbohydrates, proteins, and lipids eventually connect into glycolysis and the citric acid cycle pathways (fig 4.20)
- Carbohydrates – first choice of cellular respiration
- Lipids and Proteins – have different entry points into cellular respiration



- Glycogen from the liver and muscles, together with fats, can feed into the catabolic pathways for carbohydrates.